

Low Power AI Subsystem Overview

80-79812-252 Rev. AA

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Revision History

Revision	Date	Description
AA	December 2025	Initial release.

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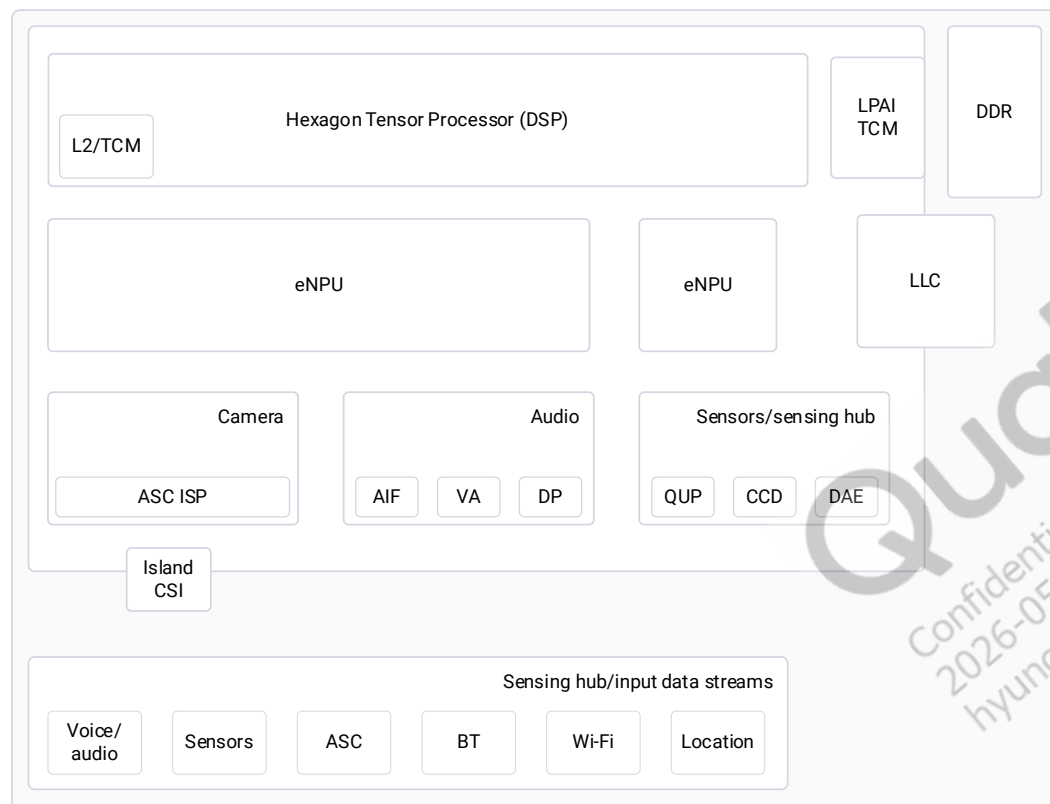


Section 1

Low Power AI (LPAI) subsystem

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LP AI architecture: SoC within SoC with multiple integrated IPs



- LP AI enables ultra low power capturing and processing of audio, voice, sensor, sensing hub, and always-on sensing camera data.
 - LP AI operates in an 'island' with the rest of the chip in a low power state (low power island) such as CXO shutdown.
- LP AI uses local embedded AI processing through multiple eNPU's
 - The eNPU's are programmable with the Qualcomm® AI Runtime (QAI RT) SDK.
 - The LP AI subsystem offloads machine learning algorithms from the DSP to the eNPU.

LP AI: Always on

- **Voice:**
 - Voice activation, voice UI noise suppression
- **Audio Context:**
 - Environments, events, music, proximity detection
- **Sensors and sensing hub**
 - Activity recognition
 - Multi-device experience
 - Presence detection

LP AI: Low power and low latency

- **Voice call**
 - ECNS
- **Audio**
 - Pre/postprocessing

LPAL architecture: SoC within SoC with multiple integrated IPs (cont.)

■ Qualcomm® Hexagon™ DSP

- A multi-threaded and VLIW processor

■ LPAL subsystem tightly-coupled memory (TCM)

- Primarily used in island use cases
- In non-island use cases, used to avoid data transactions to DDR

■ eNPU

- A big and small core provide ML offload from the DSP

■ Camera

- Always sensing camera (ASC) ISP hardware with the Qualcomm® Sensing Hub (QSH) camera framework

■ Sensor

- Provides support to control QUP
- Context change detector (CCD): Configurable hardware ML frontend

■ Audio

- Provides audio interfaces, such as SWR/TDM
- Provides acoustic activity detection hardware (HWAAD) used for wakeword and ACD

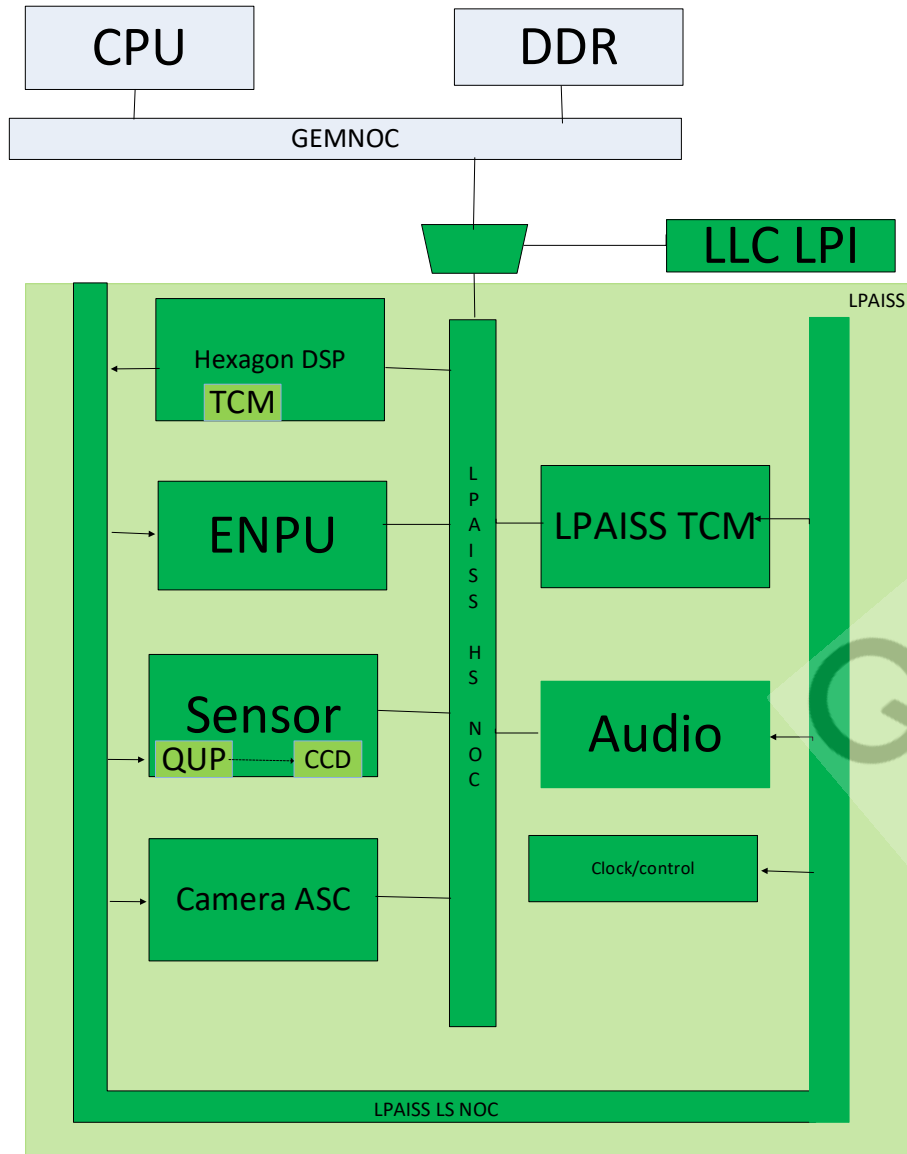
■ LLC

- A system cache that can be repurposed as island memory when the application processor is in a sleep state

■ DDR

- LPAL subsystem hardware, such as the DSP and eNPU, cold access DDR with the SMMU configured by the HLOS framework

Bus connections



- Data transfer between the DSP and DDR is done through LPAI subsystem high-speed NOC
- LPAI subsystem low-speed NOC is used to control hardware blocks, such as programming the TLMM registers
- The green hardware blocks will be power source from LPI MX/CX
- In non-LPAI mode with DDR access, chip (CHIP CX, CHIP MX) is awake



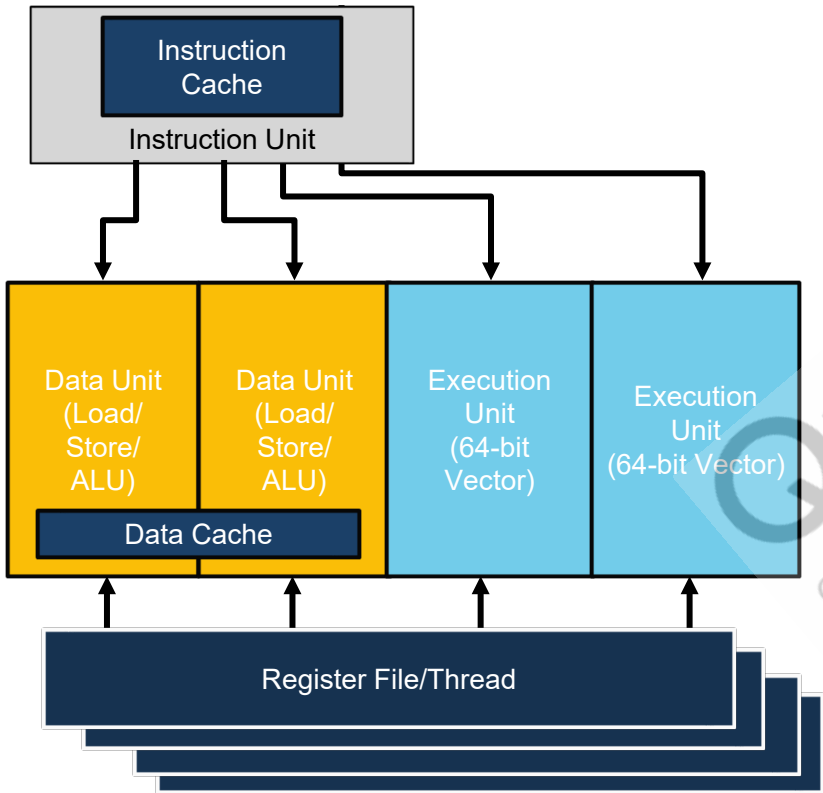
Section 2

DSP overview

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Hexagon DSP architecture

- Hexagon is a multi-threaded processor designed for LPAI use cases.
 - LPAI DSP supports multiple tech domains, such as charger, sensor, audio/voice, and ASC.
 - Up to 4 instructions can be sent in one packet



```
/* - len is multiple of 8 */
26 {
27   P0 = CMP.GT(R1,#0)          // numM > 0 ?
28   R12.L = #4                 // -4
29   R14.L = #0xFFFC            // -2
30   R15.L = #0xFFFE
31 }
P:B057CA20 75414000712C40... cod7p35_srchMaxIdxs: P0=cmp.gt(R1,#0x0); R12.L=#0x4; R14.L=#0xFFFC; R15.L=#0xFFFE;
32 {
33   IF (P0) JUMPR R31           // -3
34   R14.H = #0xFFFF            // R15:14 = (3|2|1|0) - 4
35   R15.H = #0xFFFF            // LC0 = len/8
36   R2 = ASR(R2,#3)             // R2=asr(R2,#0x3); if !P0 jumpr LR; R14.H=#0xFFFFD; R15.H=#0xFFFF;
37 }
P:B057CA30 8C024302537F40...
38 {
39   R12.H = #4                 // R12 = 4|4
40   R28 = #-32768
41   LOOP1(.cod7p35_srchMaxIdxs_outerLOOP, R1) // setup outer loop
42   P3 = CMP.GT(R2,R2)          // clean P3
43 }
44 }
```

B::Register											
R0	0	R8	0	R16	F0A32278	R24	F0305D39	SP	F0514108		
R1	0	R9	0	R17	FFFFFFFF	R25	F03D125C		A0A796BA		
R2	0	R10	0	R18	F0306028	R26	F0306032	+04	F0305D39		
R3	FFFFFFFF	R11	0	R19	F0306052	R27	F0447794	+08	00000133		
R4	0	R12	0	R20	F0305F8D	R28	F0038F30	+0C	00000000		
R5	0	R13	0	R21	F030605C	SP	F05140A0	+10	00000000		
R6	FFFFFFFF	R14	0	R22	F0305F18	FP	F05140A0	+14	00000001		
R7	FFFFFFFF	R15	0	R23	F030603C	LR	F0162E34	+18	00000000		
								+1C	00000003		
LC0	0	SA0	0	MO	0	PC	F0045920	+20	F05140F0		
LC1	0	SA1	0	M1	0	UGP	F0514120	+24	0877E893		
P	0	CS0	0	USR	00091C00	GP	F0A32000	+28	00000000		
		CS1	0	FKEY	50B1888E	FLMT	F0461080	+2C	00000001		
								+30	00000000		
STID	FFFE	ELR	F0045920	SGP0	FE697E40	BADV0	0	+34	00000000		
HTID	0	SSR	038F0027	SGP1	0	BADV1	0	+38	F0514120		
IMASK	0	CCR	0	I E U ASID	0	BADVA	0	+3C	F0162ABC		
								+40	00000000		
GELR	0	GSR	0	GOSP	0	GBADVA	0	+44	00000000		
								+48	00000000		

Example DSP Clock plan

DSP	v79* (4x MAC , 4T/2C)
Turbo	1593
Nominal	1305
SVS	864
LowSVS	614
MinSVS/LowSVS_D1	N/A
* v79 is used as an example.	

- The final clock plan will be selected based on the aggregated MPPS voting to the MMPPM service
- The static PD framework, such as SPF/SEE, will provide a framework level to vote for DSP MPPS
- See the chipset-specific clock documentation for additional details.

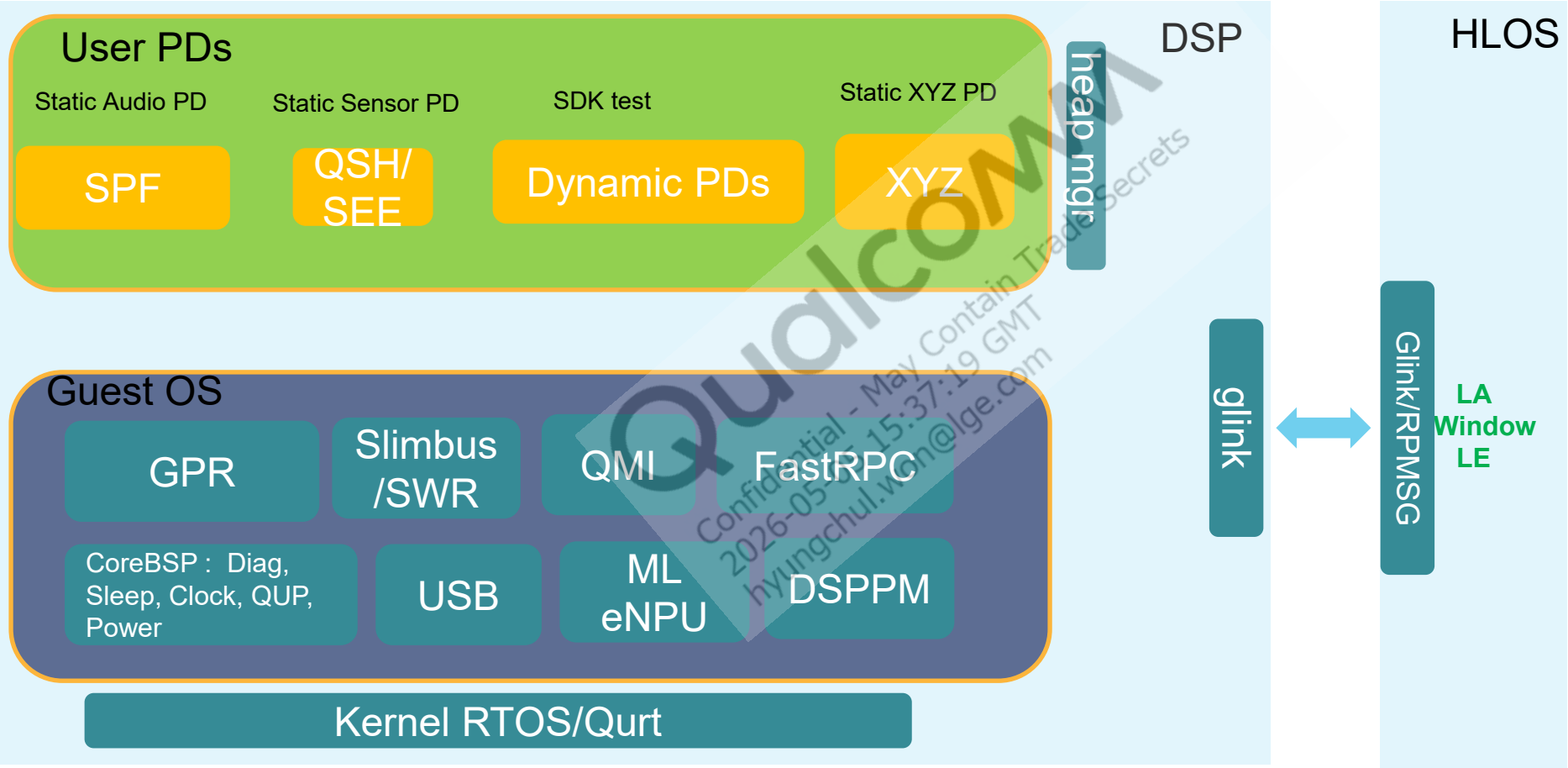
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Protection domains

- Since LPAI DSP is a real-time processor, whose stability critically affects the overall user experience, different protection domains (PDs) exist in the corresponding software architecture.
- The aDSP contains the following PDs:
 - Kernel mode
 - Access all system registers and all physical memory
 - Kernel runs in kernel mode: Non-preemptible with small memory foot-print
 - Guest OS mode
 - Access limited system registers memory mapped with OS or User mode
 - QTI drivers, most RTOS services such as memory, process management
 - User mode
 - OEM programs, user space drivers, and RTOS user libraries

LPAI DSP software components

- HLOS as client: Creates/triggers the use case.
- Static PD: Provides the service/framework.



Major software components

■ QURT OS

- Real-time operating system (RTOS) that handles thread scheduling and priority.

■ Core services

- Handles memory heap, interprocessor communications (IPC), diagnostics, power (NPA), and sleep functions.

■ aDSP power management (PM)

- PM module for LPAI DSP.

■ Sensors Execution Environment (SEE)

- Framework provided to develop custom sensor algorithms.

■ Qualcomm Sensor Hub (QSH)

- Enables low power contextual data processing for Bluetooth Low Energy (BLE), Wi-Fi, audio, and camera use cases.

■ Audio module database (AMDB)

- Provides dynamic module support.

■ Fast RPC

- Remote HLOS heap.
- Dynamic module loading and remote procedure calls (RPC) for offloading computations to the DSP.

■ Generic packet router (GPR)

- Provides a wrapper for audio/voice IPC

■ Signal processing framework (SPF)

- Framework used for audio and voice solutions.

■ Battery management

- PMIC charging and fuel gauge drivers.

■ Heap manager

- Manages the heap (island and non-island) per PD.

TCM configuration

- Size of the total TCM is chipset specific
- The *adsp_proc/config/target.adsp/cust_config.xml* file provides the interface to adjust the TCM for each memory pool of each function.

```
<physical_pool name="AUDIO_ISLAND_LPASS_TCM_POOL" island="true" collapse="tcm">  
  <region base="0x06000000" size="0xF0000" cache_policy="l1_wb_l2_cacheable"/>  
</physical_pool>  
<physical_pool name="CAM_ISLAND_LPASS_TCM_POOL" island="true" collapse="tcm">  
  <region base="0x060F0000" size="0x8F000" cache_policy="l1_wb_l2_cacheable"/>  
</physical_pool>
```

Island memory available for customization

- Check the build log under *adsp_proc\build\ms*.
- OEMs can remove QTI static island code and data.

***** Q6-ADSP Memory Footprint *****

AUDIOPD Heap:
UKERNELPD Image:
Alignment/Qurt Allocation:
Total Image Size Consumed:
Available Size: 3.844MB
Island Size in ROOTPD:
Island Size in SENSORPD:
Island Size in AUDIOPD:
Island Size in UKERNELPD:
Total island Size Consumed:
Available Size in Island: 2.156MB
L2 Size:
Q6 TCM Size:
LPASS TCM Size:

physpool	Used Size (KB)	Free Size (KB)	Total Size (KB)
AUDIO_ISLAND_LPASS_TCM_POOL	0.0	1472.0	1472
AUDIO_ISLAND_TCM_PHYSPOOL	464.0	624.0	1088
CAM_ISLAND_LPASS_TCM_POOL	0.0	572.0	572
QSHTECH_ISLAND_POOL	228.0	1820.0	2048
QSH_ISLAND_POOL	460.0	52.0	512
QURTOS_ISLAND_POOL	512.0	0.0	512
SSC_ISLAND_POOL	464.0	560.0	1024
TCM_PHYSPOOL	132.0	40.0	172
CAM_LLCC_ISLAND1_POOL	212.0	4140.0	4352



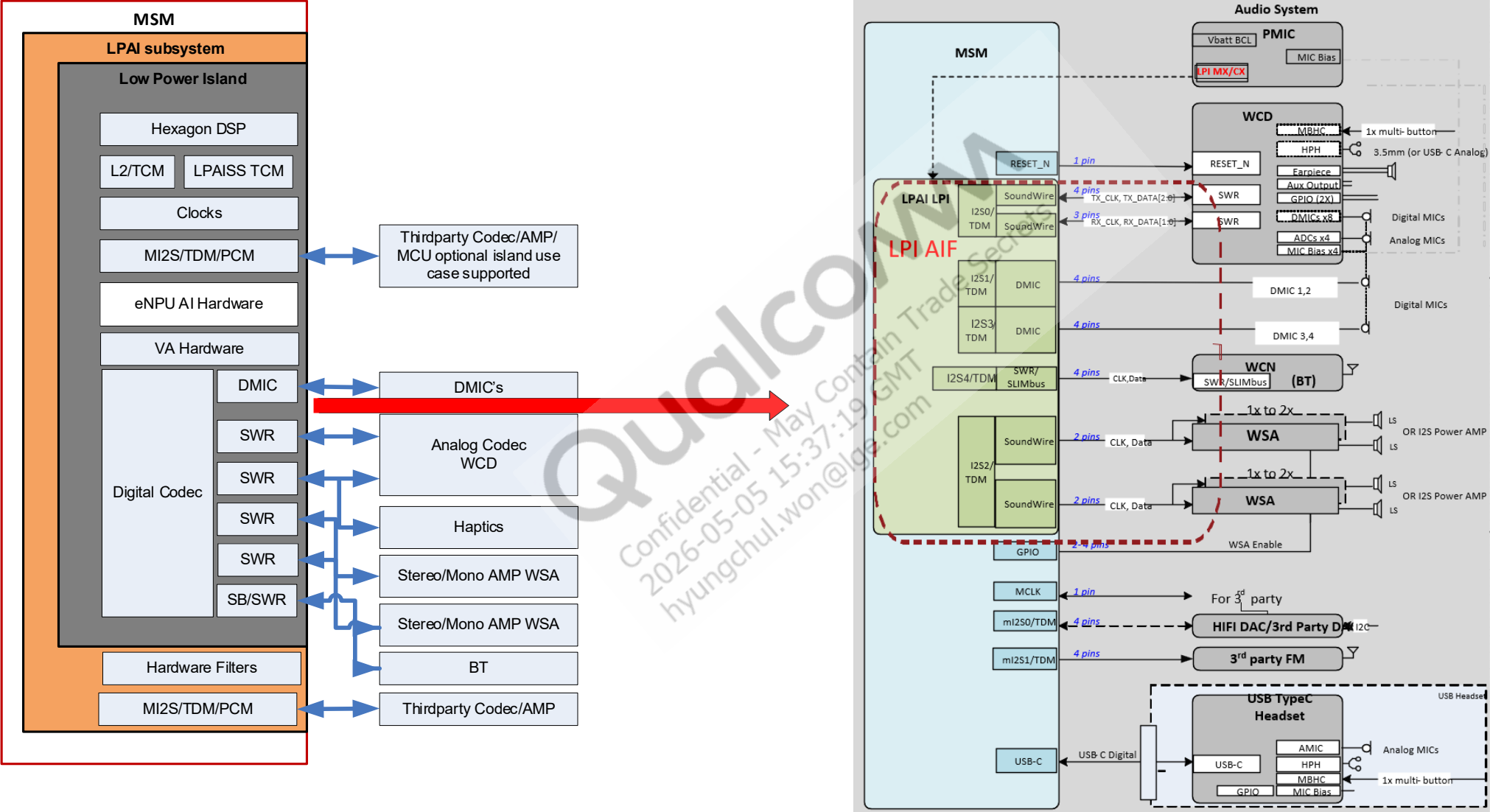
Section 3

Audio overview

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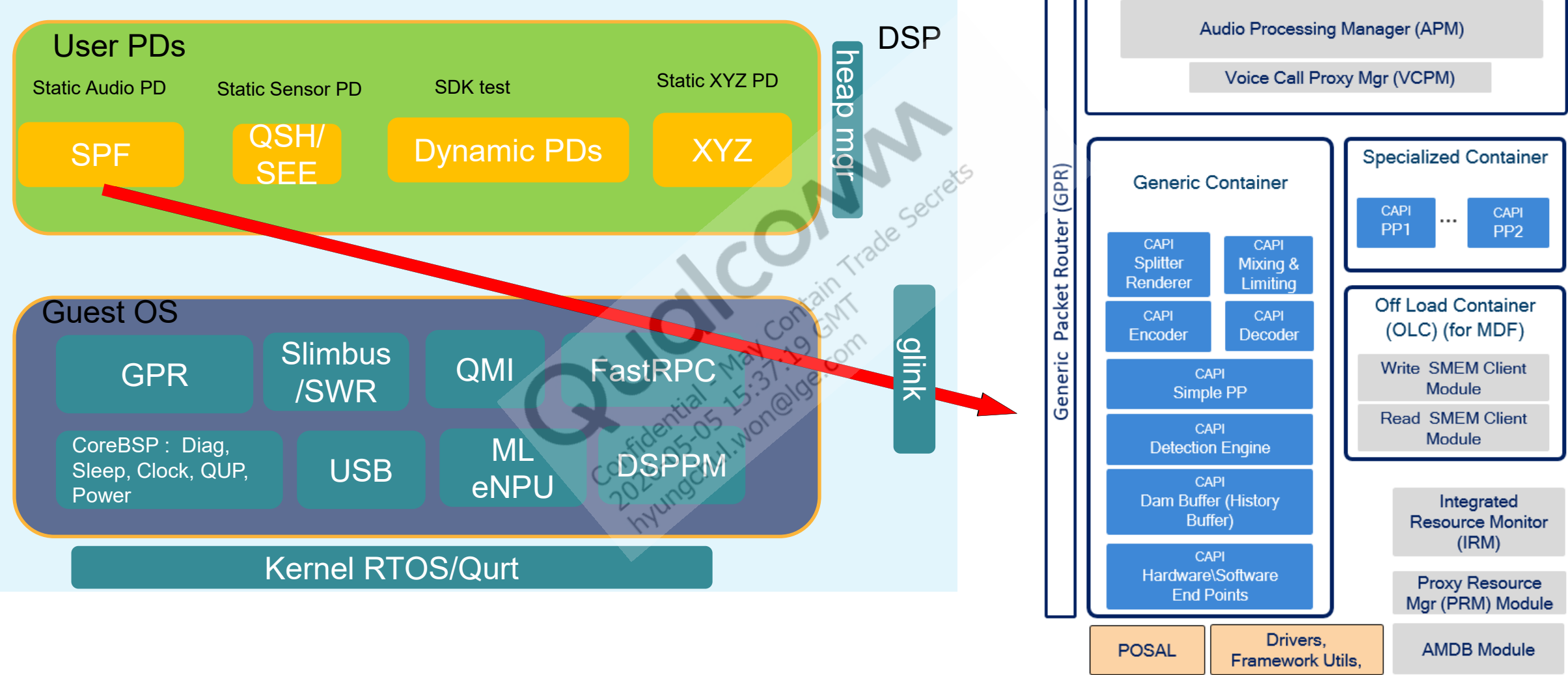
Audio hardware components and interface

Audio IOs



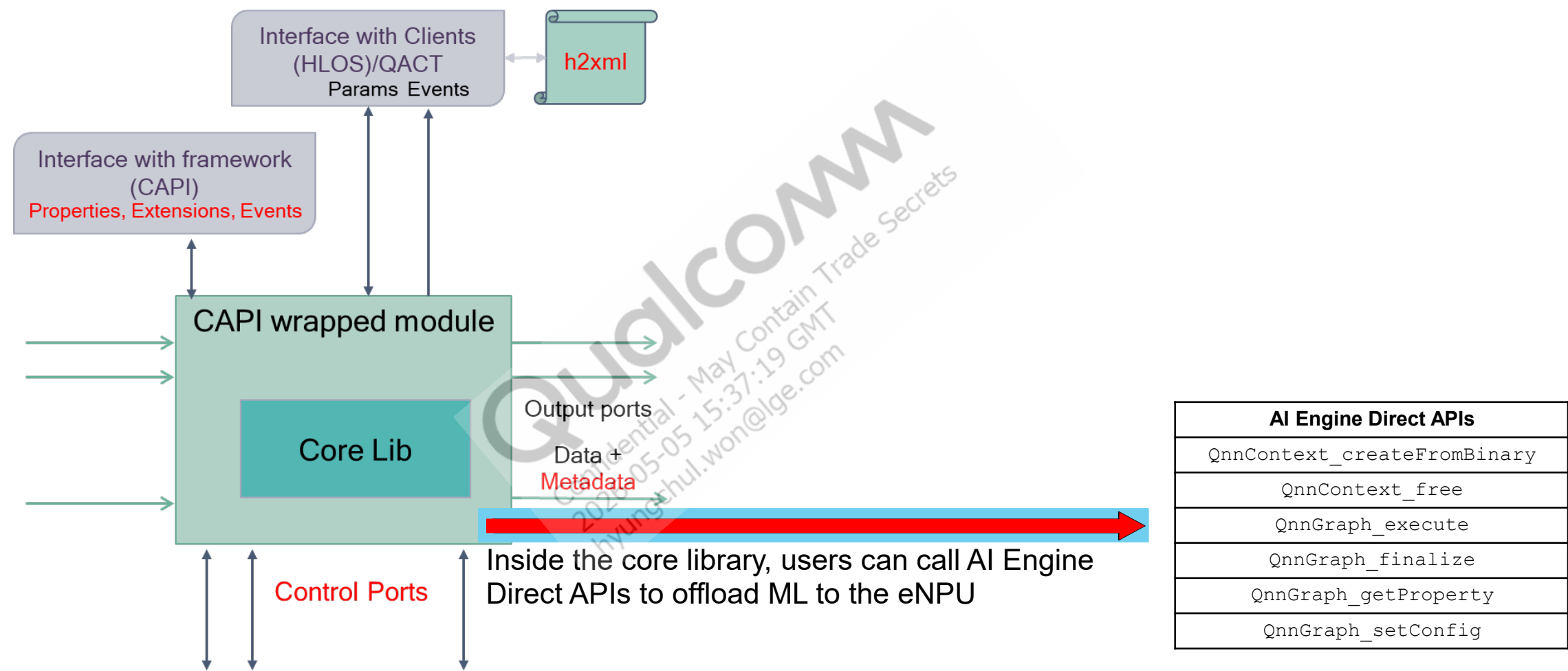
Audio software components

Signal Process Framework

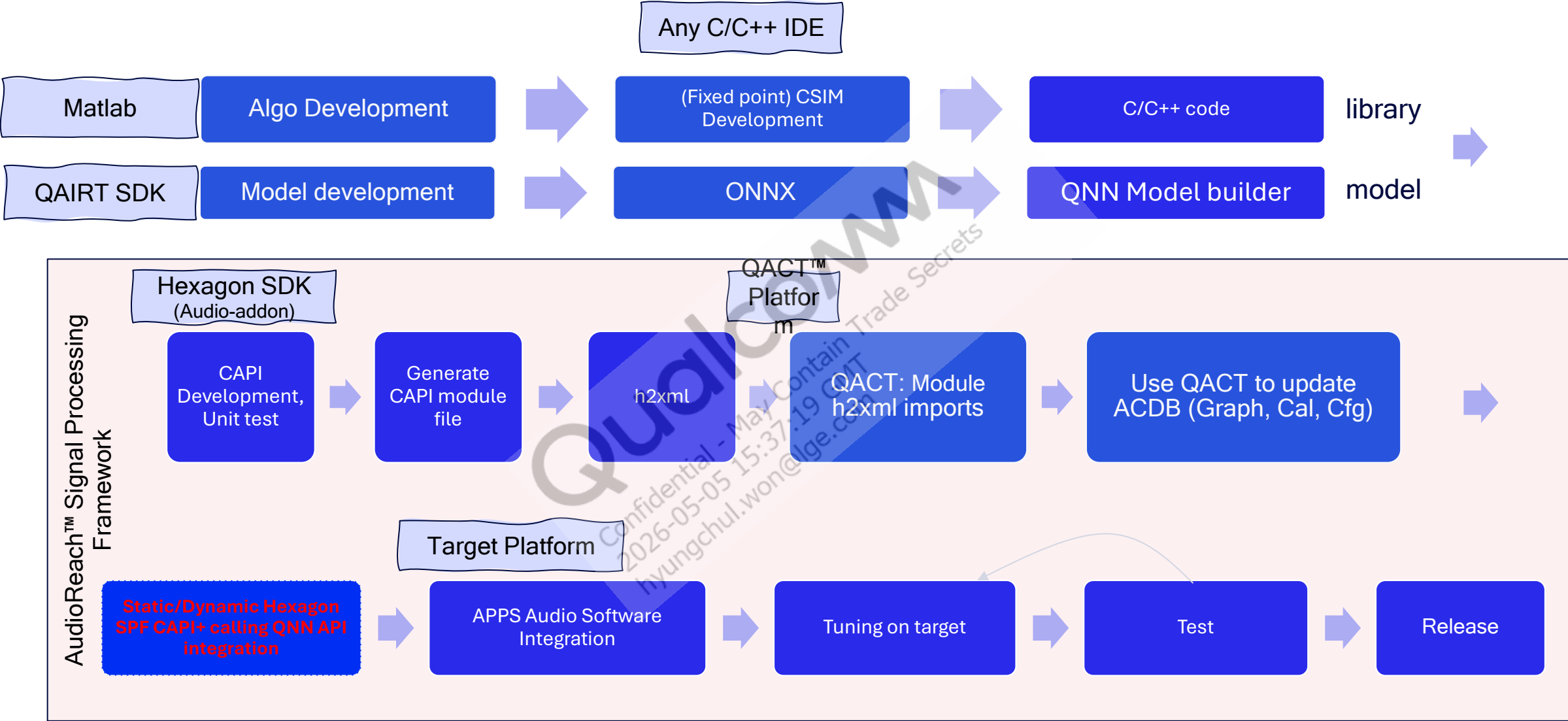


Common audio processing interface (CAPI) version 3 (v3)

- CAPI provides a common interface to handle the audio PCM (or compressed) data provided by SPF.



Typical development workflow



SPF CAPI sample code

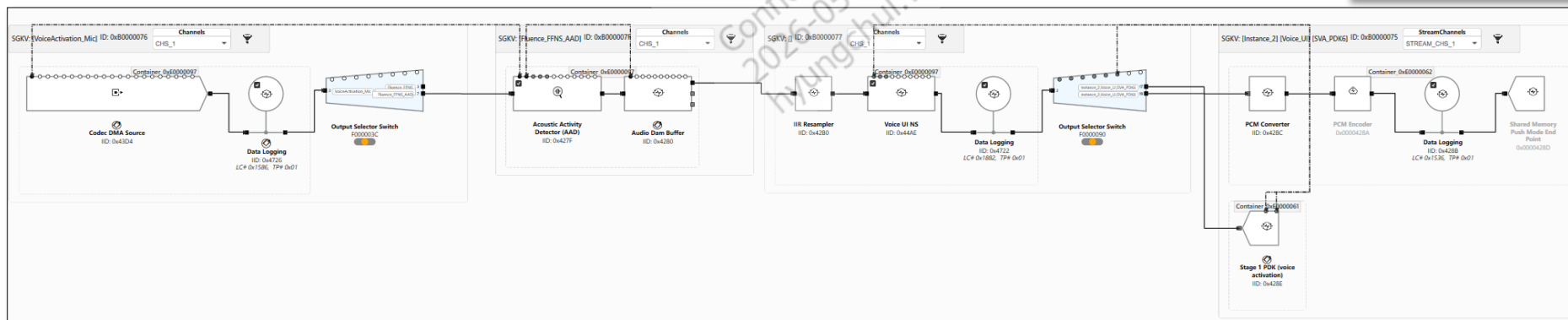
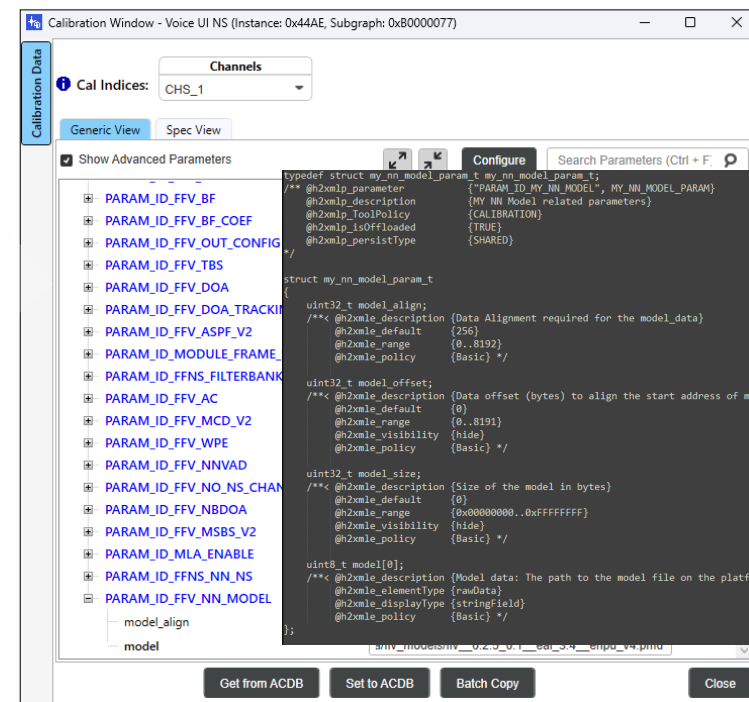
■ Hexagon SDK and audio add-on

addons\audio\spf\examples

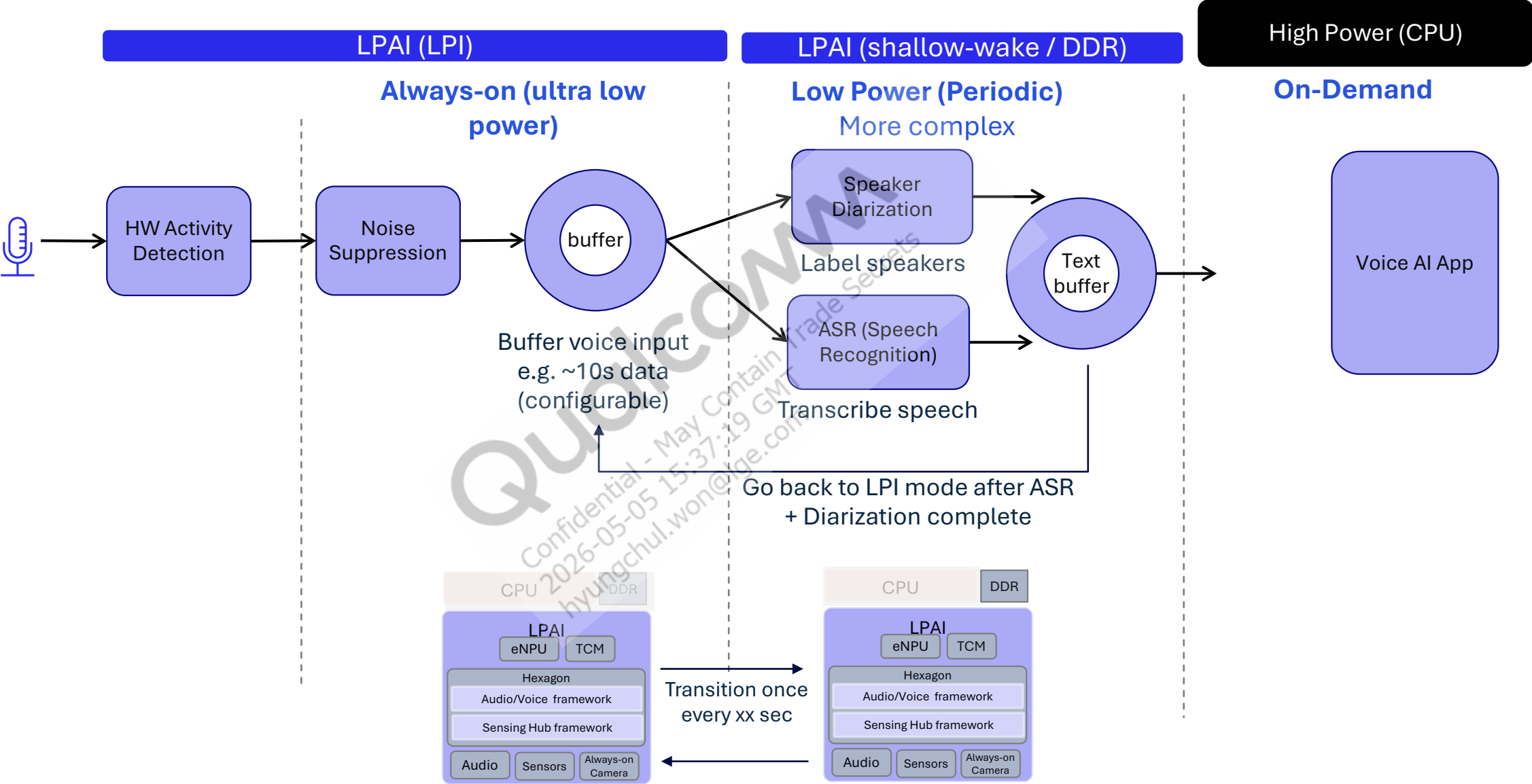
```
|—bt_decoder\  
|—bt_encoder\           // Codec integration  
|—capi_decimate\  
|—capi_decimate_qurt_based\ // PCM Post processing integration  
|—capi_detection_eai\  
|—capi_detection_engine\  
|—capi_detection_engine_LLC\ // Wakeword integration  
|—capi_detection_qnn\  
|—capi_detection_qnn_htp\  
|—capi_ecns\           // Voice ECNS integration  
|—capi_gain\  
|—capi_gain_cpp\  
|—capi_gain_multithreading\  
|—capi_multi_port\  
|—capi_passthru\  
|—capi_sp\             // third-party party speaker protection algo integration  
|—capi_spf\  
|—so_version\  
|—upd\                 // UltraSound proxy module detection
```

Graph customization using QACT

- Open product ACDB in QACT
- Filter the required graph
- Discovery Wizard -> Import h2xml of the new module
 - imports module-id & param-id. Manually add file name and interface type in Module Manager.
- Click “Start Graph Modification”, Insert your module and connect to the graph. Stop graph modification and save the ACDB.
- Model integration (example ways)
 - Remote file system APIs (testing only) or link the sound model to the algo code.
 - For EC/NS modules add sound model in QACT.
 - WakeOnVoice: models are pushed from HLOS driver
 - Audio Context Detection: Model files picked from resource manager XML based on application given params (e.g. context-id)
- Push ACDB, .so files, model files to target
- Run the use case and test



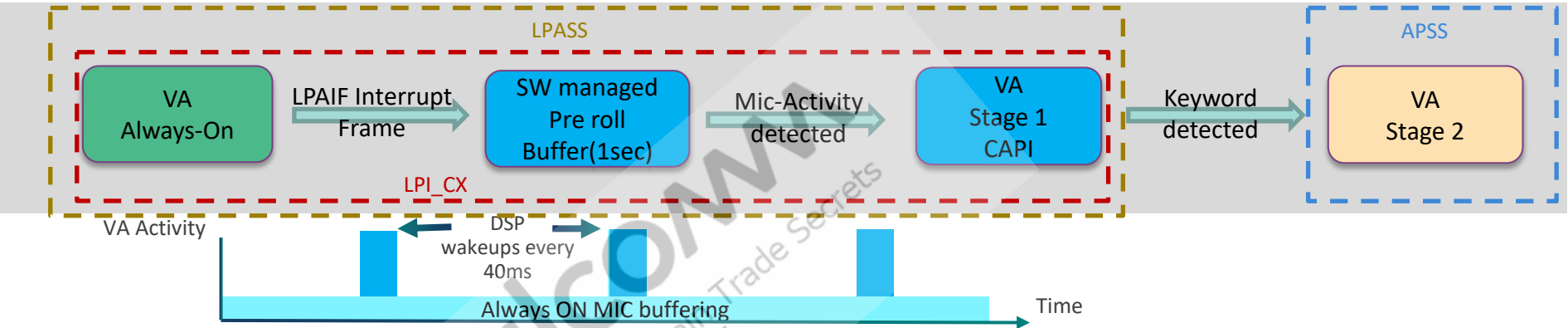
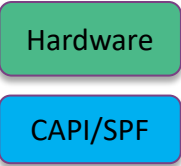
Data flow of always-on conversation recording



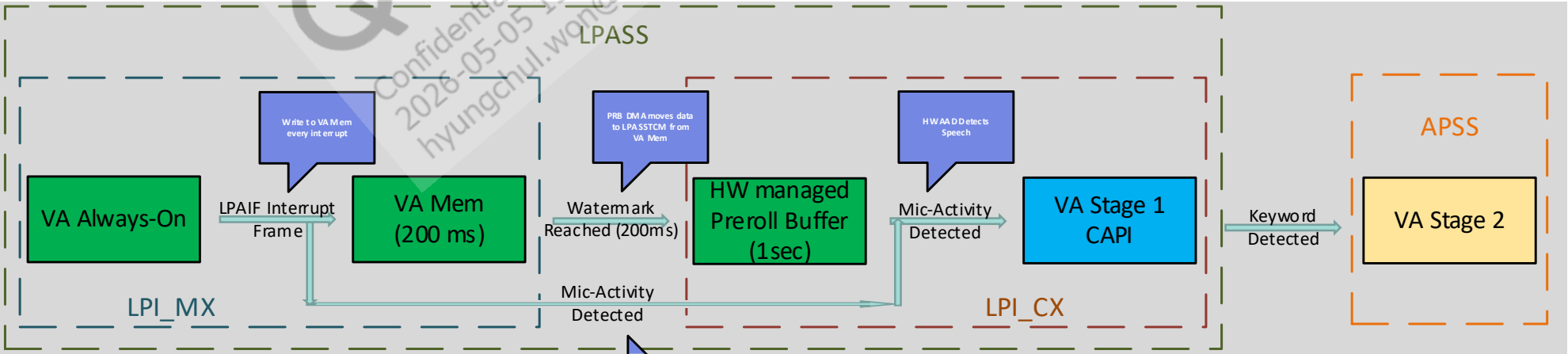
VA hardware and software blocks

- Data buffering in the hardware block and then wake up DSP

Software AAD



Hardware AAD





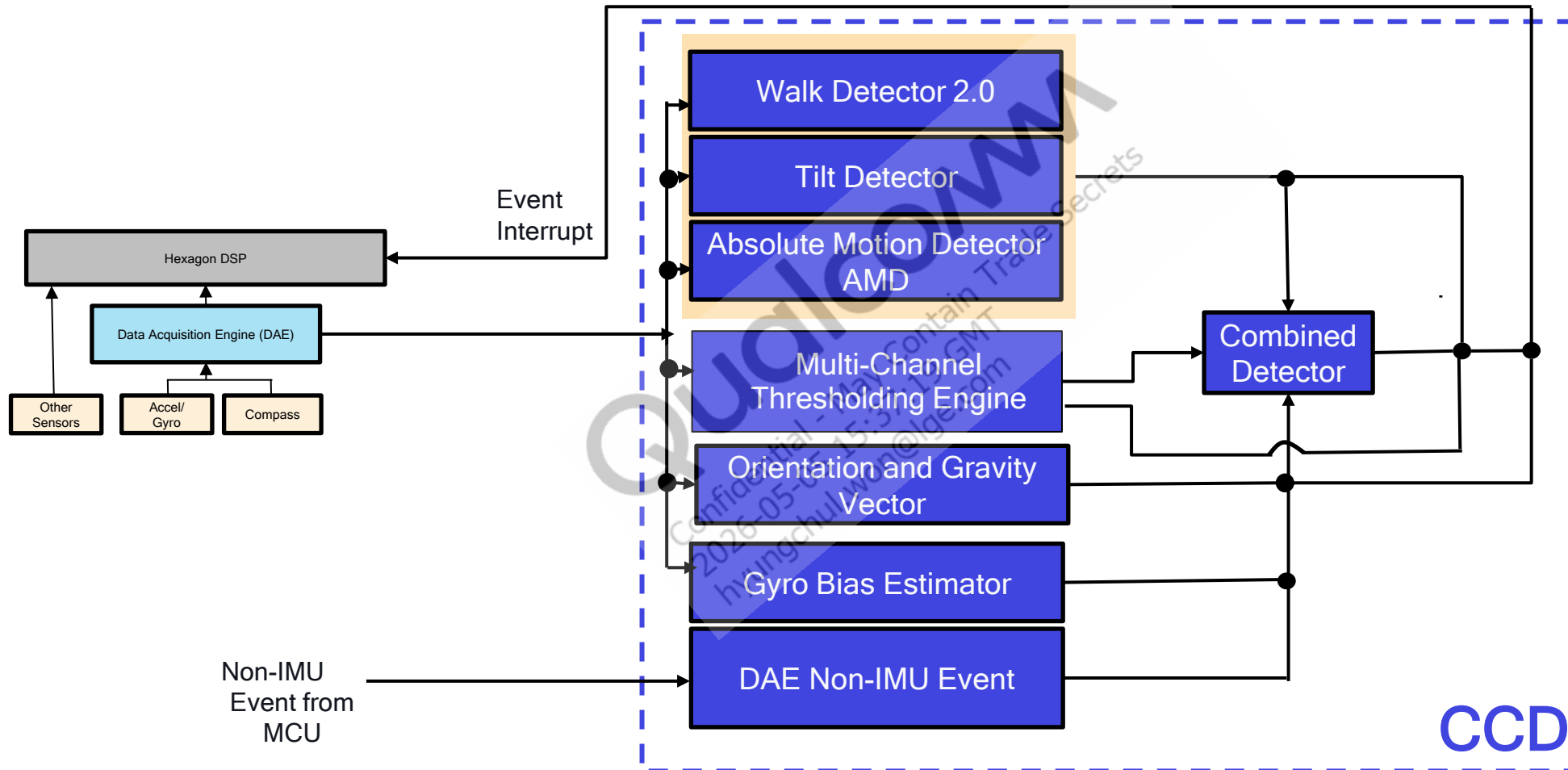
Section 4

Sensor overview

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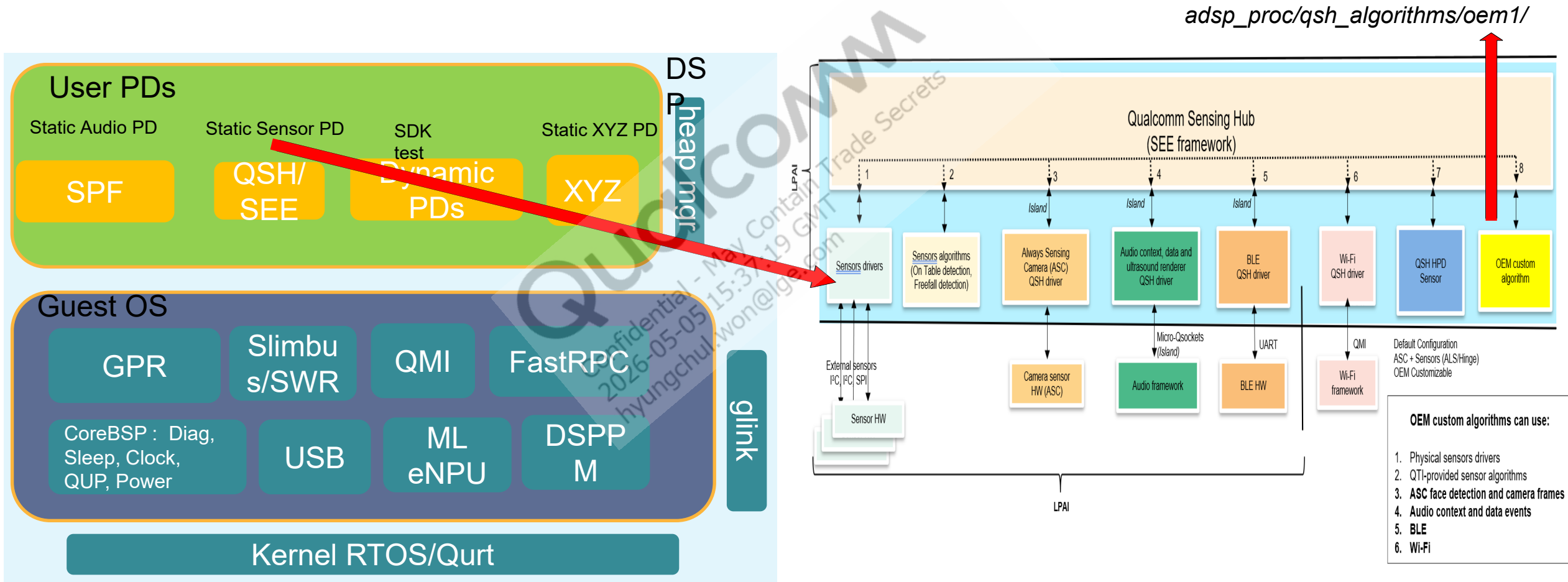
CCD architecture block diagram

- CCD is a hardware-based sensing ML frontend that reduces the need for MCU and DSP wake ups.
- See *Context Change Detector (CCD) 6.0 and Context Engine Configuration Tool (CECT) 6.0 User Guide* (80-60672-3) for more information.



Sensor software components

- Qualcomm Sensing Hub (QSH) drivers
- HLOS sensor framework to request sensor data and events
- QSH wakes up HLOS sensor framework to handle events and data, such as gyroscope data



Qualcomm universal peripheral

- Configuration is done through JSON configuration files.

target_xyz_0.json

```
'bus_type'           0    // I2C
'max_bus_speed_khz'  400
'vddio_rail'         'L1B_E0'
'vdd_rail'           'L18B_E0'
```

- **SSC_QUP**

- Hardware IP inside LPAI subsystem
- Only accessible from LPAI DSP software
- Can only access PRAM (no DDR)
- Faster than TOP QUP
- Suitable for island use cases

- **TOP QUP**

- Resides outside of LPAI subsystem
- Accessible by LPAI DSP and HLOS software
- Can only access DDR
- Slower than SSC QUP
- Not suitable for island use cases

```
owner : ipszzai ,
"bus_type":{ "type": "int", "ver": "0",
  "data": "0"
},
"bus_instance":{ "type": "int", "ver": "0",
  "data": "2"
},
"slave_config":{ "type": "int", "ver": "0",
  "data": "92"
},
"i3c_address":{ "type": "int", "ver": "0",
  "data": "40"
},
"min_bus_speed_khz":{ "type": "int", "ver": "0",
  "data": "0"
},
"max_bus_speed_khz":{ "type": "int", "ver": "0",
  "data": "400"
},
"reg_addr_type":{ "type": "int", "ver": "0",
  "data": "0"
},
"dri_irq_num":{ "type": "int", "ver": "0",
  "data": "66"
},
"irq_pull_type":{ "type": "int", "ver": "0",
  "data": "0"
}
```



Section 5

eNPU overview

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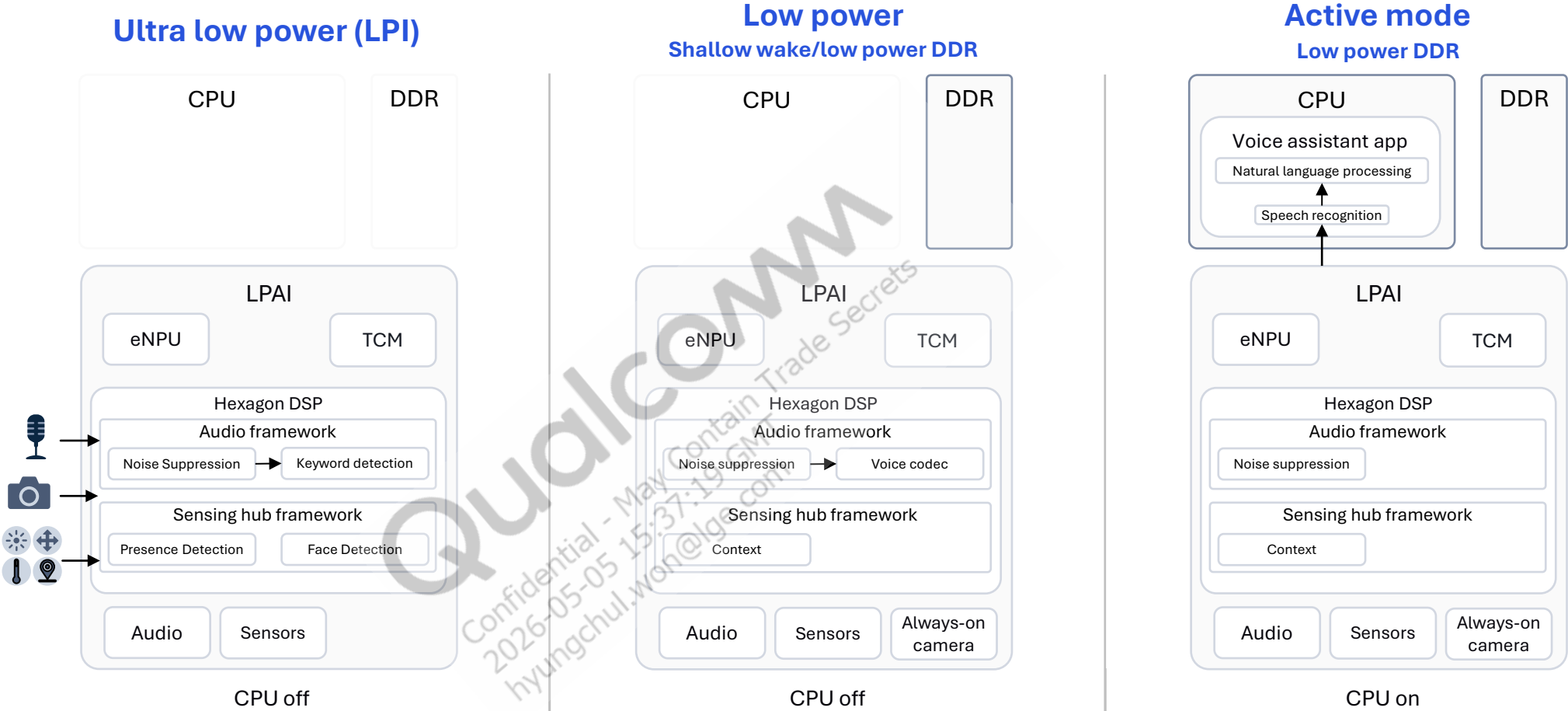
What is eNPU

- QNN framework provides frt_ratio and affinity to allow Hexagon DSP clients to offload inference to the big eNPU or boost the eNPU clock.

	Big eNPU (For example 1024 MAC)	Little eNPU (For example: 256 MAC)
Focus	■ Compute heavy workloads ■ CNNs	■ Memory bandwidth heavy workloads, for example: LSTM ■ Lower complexity
Use cases	Small ASR/NLP (attention/transformer)	■ Sensors, Activity Recognition ■ Voice Activation ■ Noise Suppression ■ Context Detection
Performance	2 TOPs: 1.65T (1024MAC) + 0.4T (256MAC)	
Ops support	181 ops Improvements for conformer, CNN, and attention	
Precision	+4w8a	
Compression	Weights and Activations: 60% for 80% sparse models	
Runtime + APIs	QNN (Qualcomm AI Stack)	

See [Supported Operations](#) for more information.

LPAI operation modes

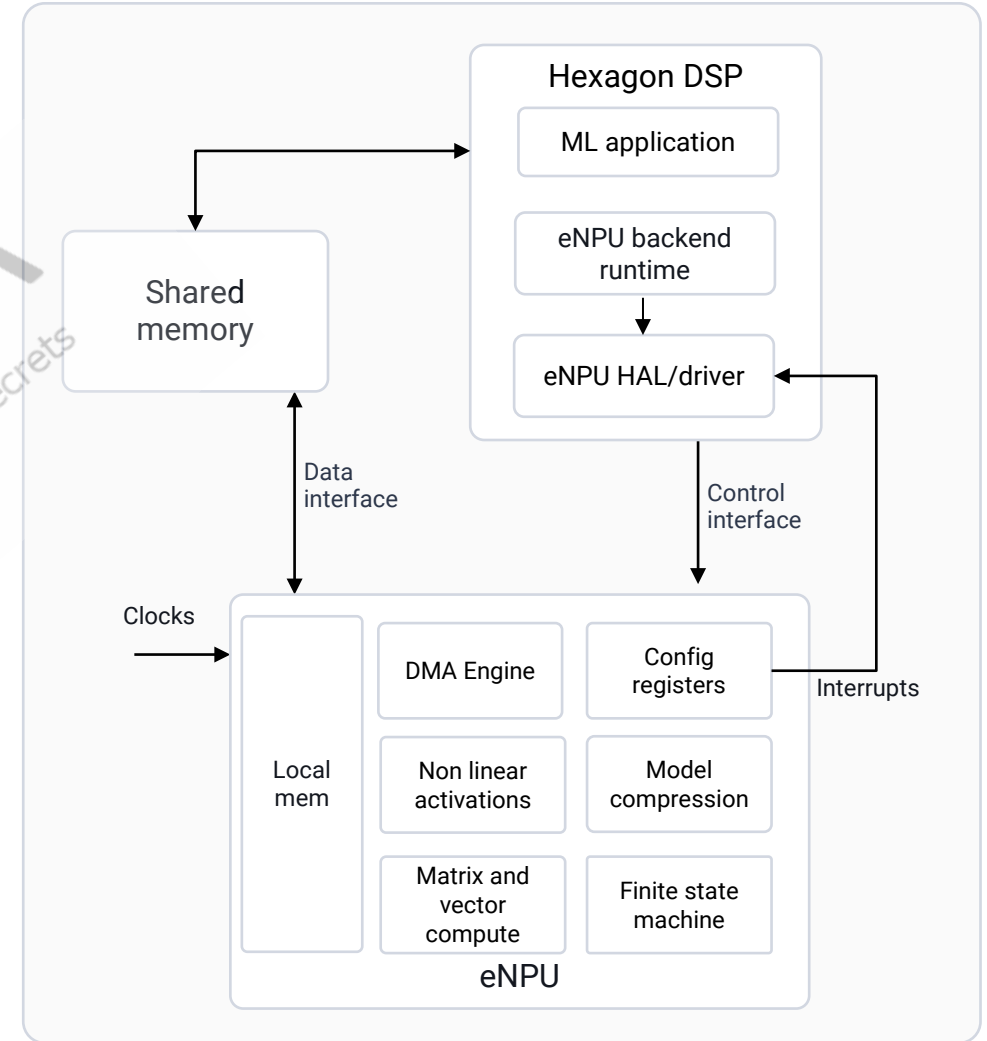


Operating Model	CPU off	CPU off	CPU on
Memory footprint	Island memory	DDR (model typically ≤ 200 MB)	DDR (model typically ≤ 200 MB)
Technologies†	Audio, Voice AI, Sensors, Sensing Hub		
Use Cases	AI runtime (QNN) + software framework (audio (AudioReach) or Qualcomm Sensing Hub (QSH))		

* Not all chipsets include LLC in LPI. † LPI technologies vary by chipset

eNPU machine learning accelerator

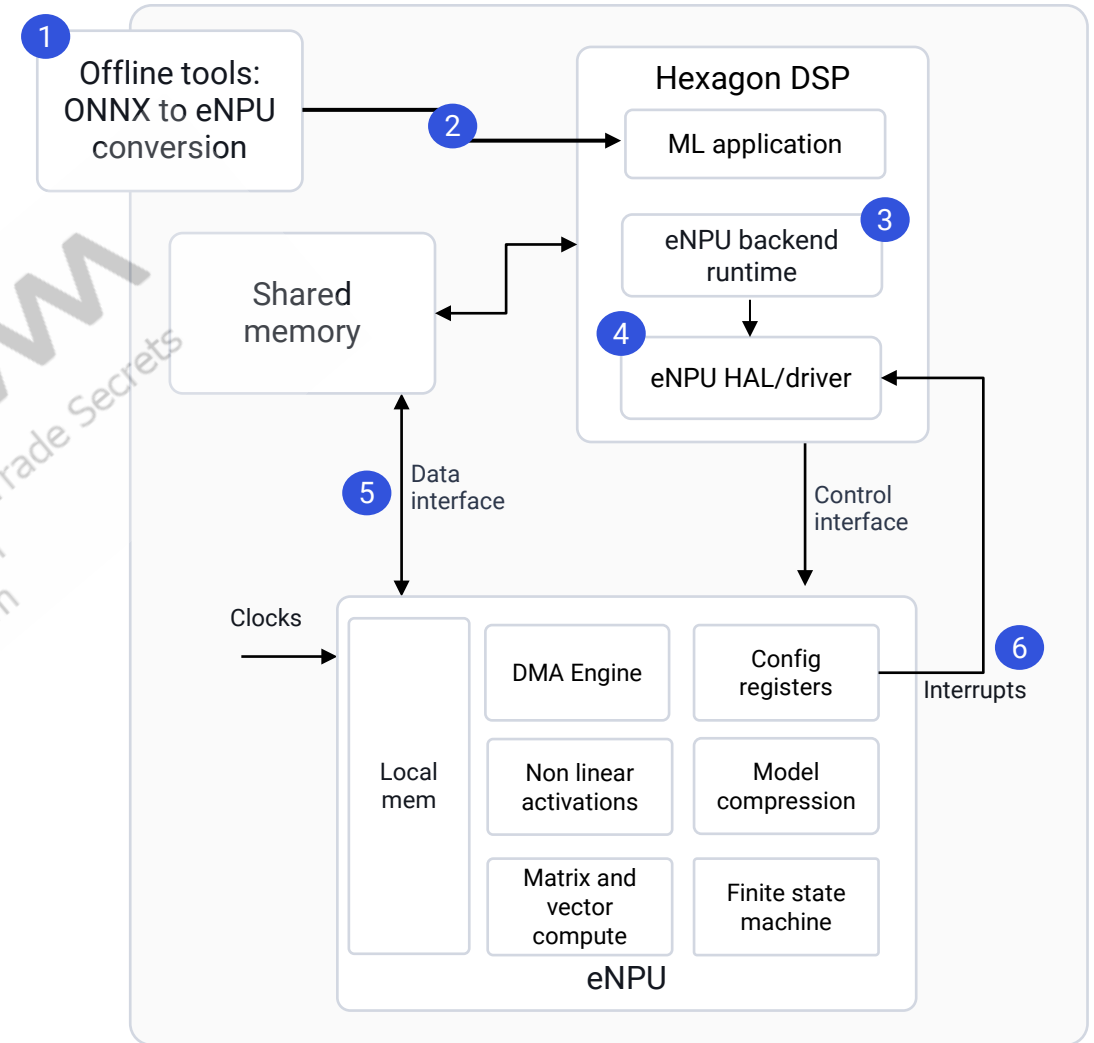
- Offloads ML processing from the DSP
 - Model execution on the eNPU is initiated within the LPAI
 - Does not require any APPS-CPU initiation or processing
- Supports commonly used operators:
 - CNN, RNN, LSTM, GRU, Attention, etc.
 - Details follow later
- Fixed-point inference engine
 - eNPU6: 8w8a, 8w16a, 16w16a , 4w8a, 4w16a
- Compression
 - Offline Weight compression (runtime decompression)
 - Runtime Sparse Activation compression



See *eNPU Design Guidelines* (80-37209-5) for information on developing eNPU-friendly model architectures.

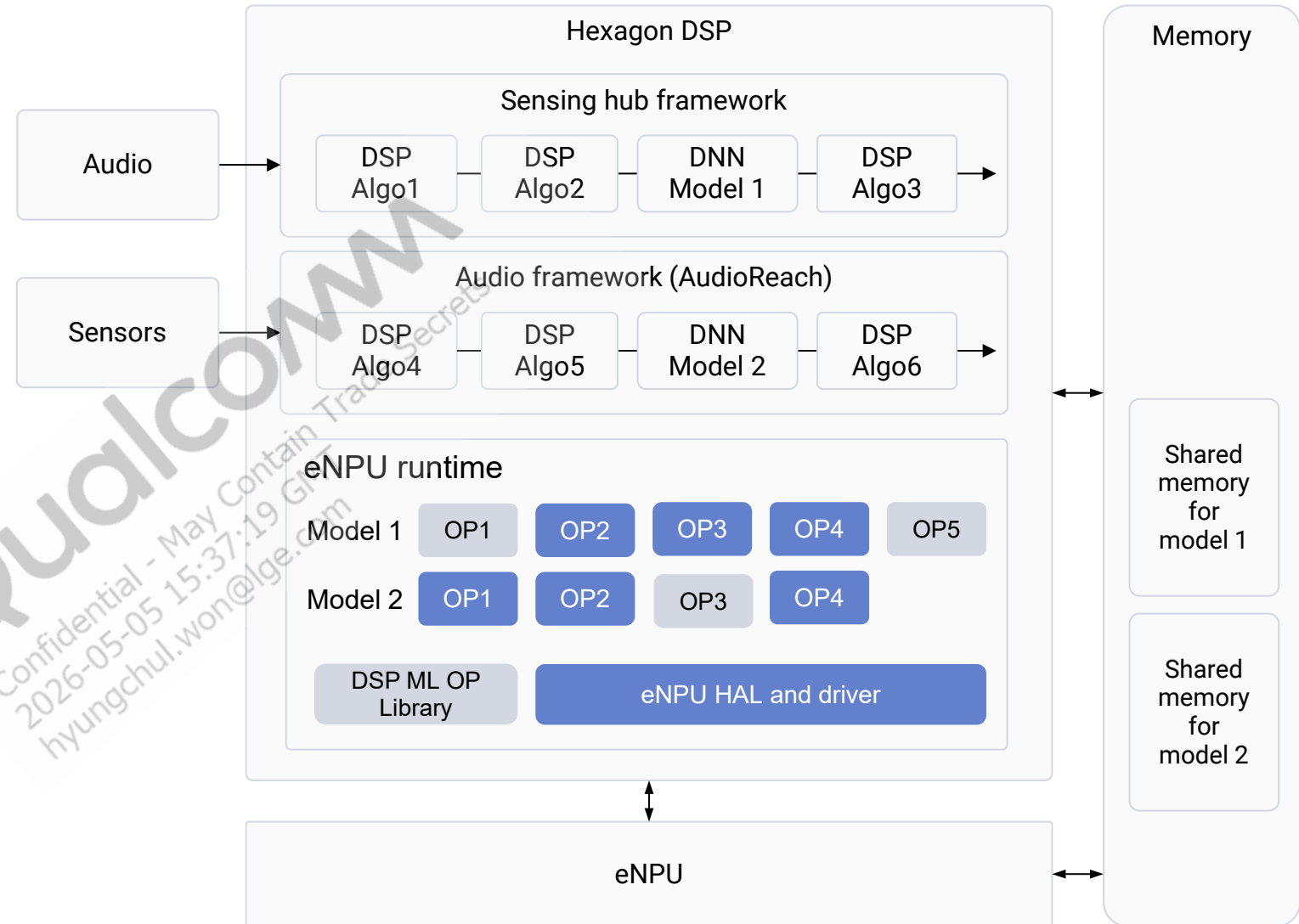
eNPU programming model

1. Create and convert offline models.
 - Create and train PyTorch/TensorFlow model offline.
 - Convert the offline models to quantized eAI models.
2. Embed the eAI model in the use case application.
3. During initialization, eNPU backend/runtime parses the model graph and creates hardware and software operation list and configures operations for eNPU.
4. During input frame processing, eNPU runtime executes the graph and invokes eNPU processing for supported operations through the driver.
5. eNPU processes each operation by reading input and weights from shared memory and generating output into shared memory.
6. An interrupt is generated after each operation or group of operations.

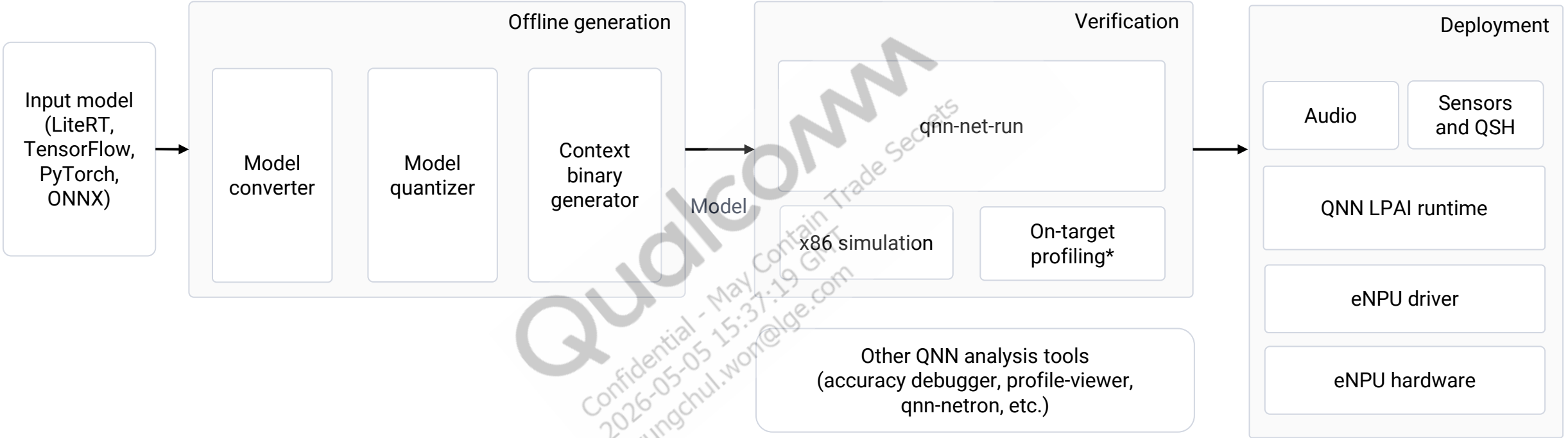


eNPU programming model (cont.)

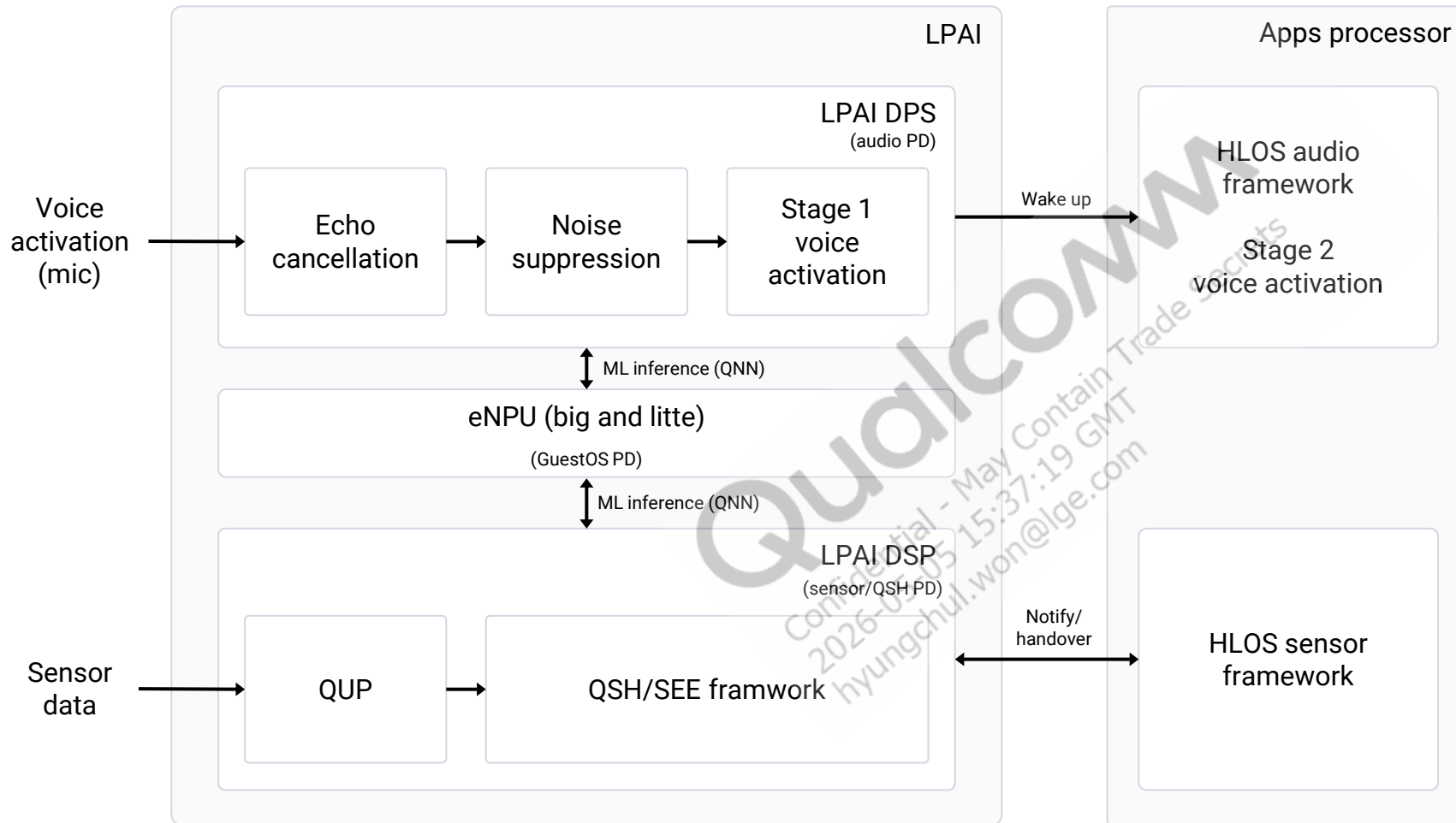
- eNPU runtime offloads operations as supported to the eNPU
- Offload happens transparently
- Support for client priority and core selection/affinity
 - Adds flexibility to control each client's priority in concurrent scenarios
 - Optimizes the workload with proper core selection and affinity settings
 - Helps the eNPU more efficiently dispatch jobs



QNN-LPAI workflow overview

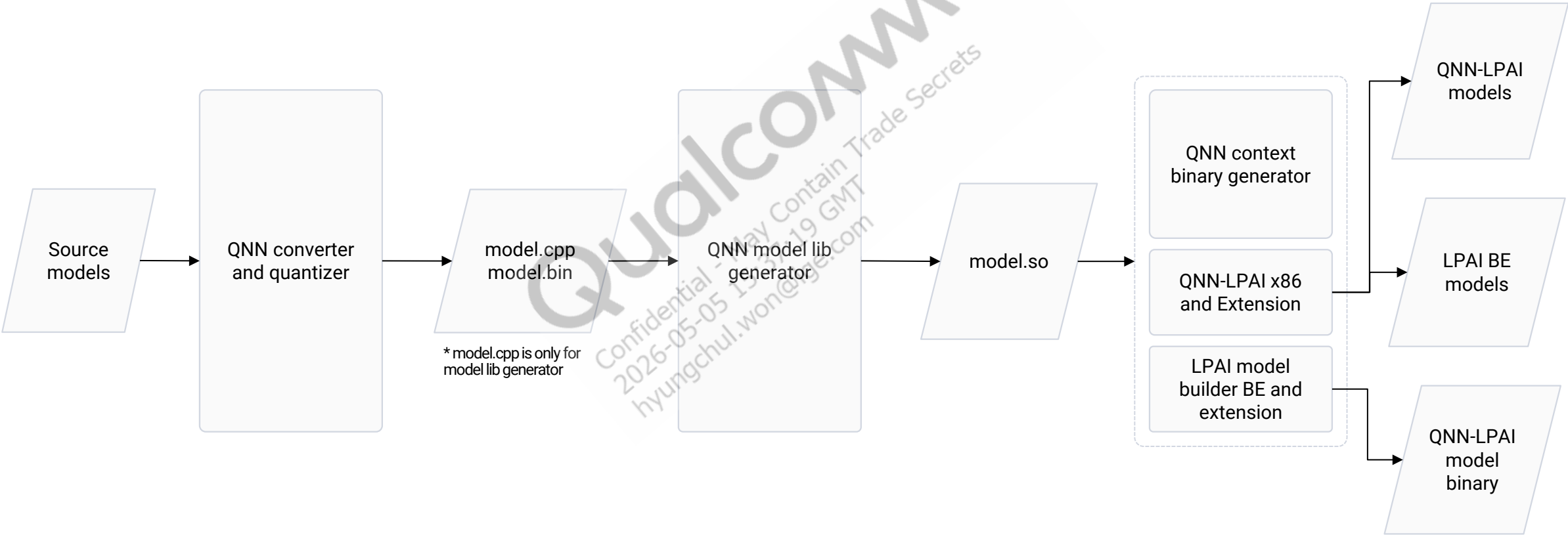


Data flow when DSP offload to eNPU

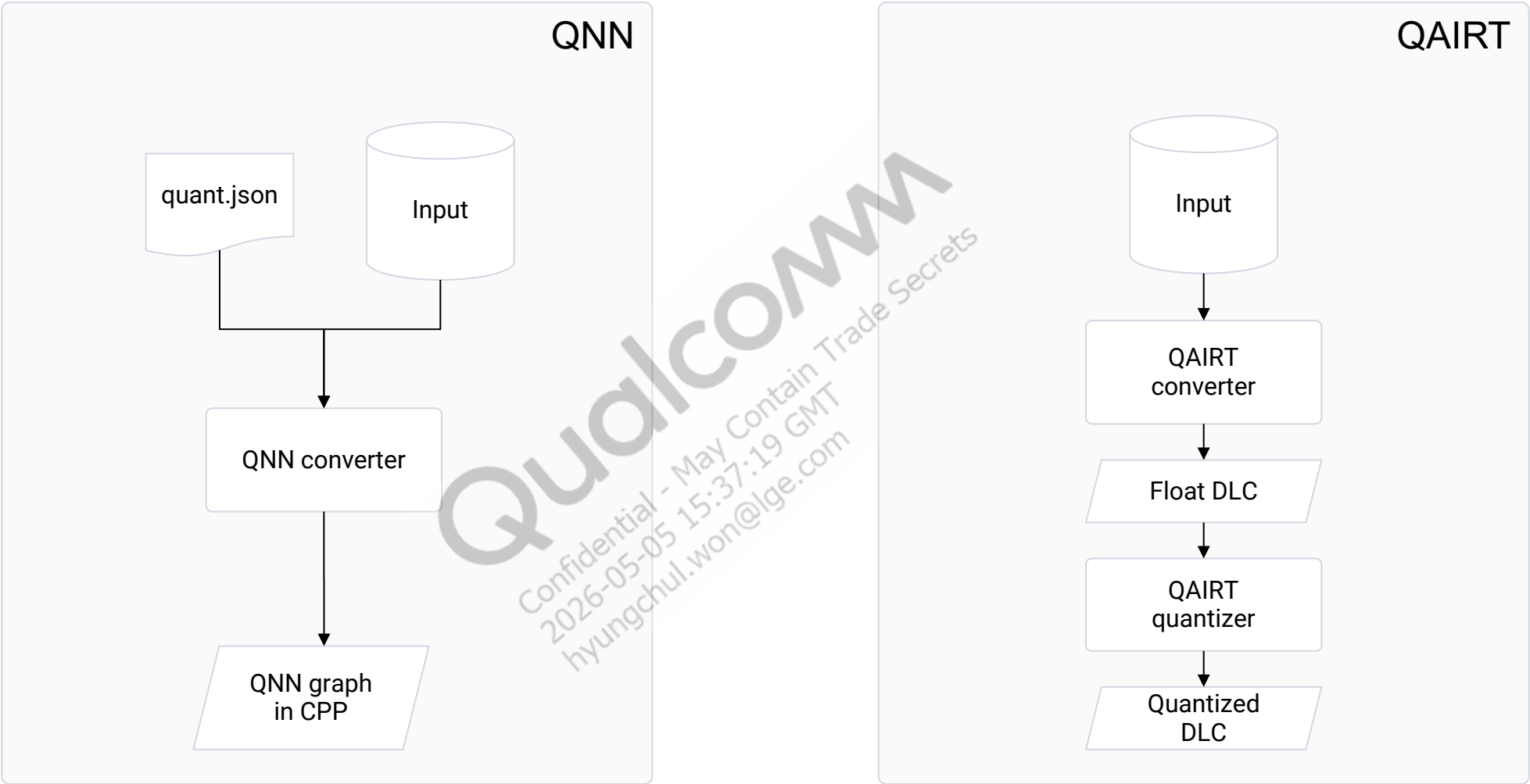


- The eNPU driver is ml-graph based and the hardware job is scheduled based on software priority. Soft affinity and FTRT are also supported to boost the eNPU clock.
- In concurrency, such as audio and sensor use case running (voice keyword detection + HPD in island mode), there will be two model to be offloaded. Each ML model will be offloaded on different eNPU core

Qualcomm AI Runtime LPAI models



Model conversion and quantization

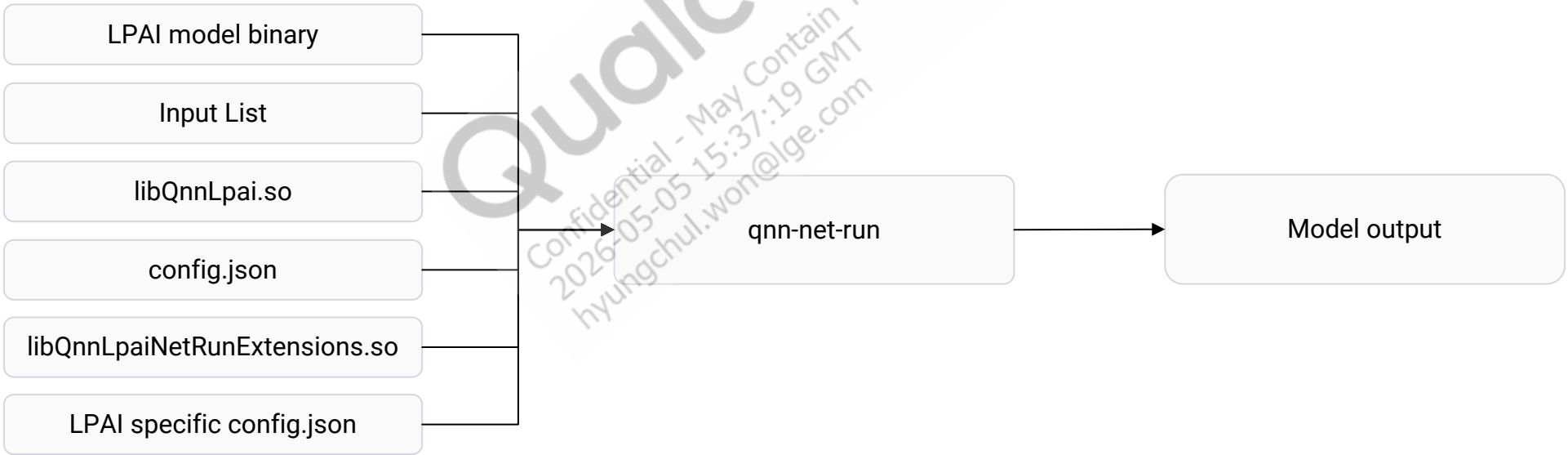


Qualcomm AI Runtime SDK tools

Tool	Description	Example usage
qnn-tensorflow-converter	Converts a model from the framework to a CPP file representing the model as a series of QNN API calls and creates a binary file containing static weights of the model.	<pre>qnn-onnx-converter --input_network crnn.onnx --quantization_overrides crnn.encodings --multi_time_steps_lstm --input_list input_list.txt</pre>
qnn-pytorch-converter		
qnn-tflite-converter		
qnn-onnx-converter		
qairt-converter	Converts a model from the framework to a DLC file representing the QNN graph format that can enable inference on Qualcomm AI hardware. The converter automatically detects the framework based on the source model extension.	<pre>qairt-converter --input_network crnn.onnx</pre>
qairt-quantizer	Converts non-quantized DLC models into quantized DLC models.	<pre>qairt-quantizer --input_dlc crnn.dlc</pre>
qnn-model-lib-generator	Compiles QNN model source code into artifacts for a specific target.	<pre>qnn-model-lib-generator -b crnn.bin -c crnn.cpp -o . -t x86_64-linux-clang</pre>
qnn-context-binary-generator	Creates a context binary using a particular backend and consuming a model library created by qnn-model-lib-generator.	<pre>qnn-context-binary-generator --model ./x86_64-linux-clang/libcrnn.so --backend \${QNN_SDK_ROOT}/lib/x86_64-linux-clang/libQnnLpai.so --binary_file crnn_lpai --config_file params.json --output_dir ./output</pre>

qnn-net-run

Tool	Description	Example usage
qnn-net-run	Consumes a model library compiled from the output of the QNN converter and runs it on the specified backend. Configuration files: ■ params.json ■ lpai_config.conf	<pre>qnn-net-run --input_list <PATH-TO-INPUT-MAPPING> --retrieve_context <PATH-TI-FINAL-CACHE-BINARY> --config_file <PATH-TO-PARAMS.JSON> --backend <PATH-TO-LPAI-BACKEND></pre>



qnn-net-run: Configuration files

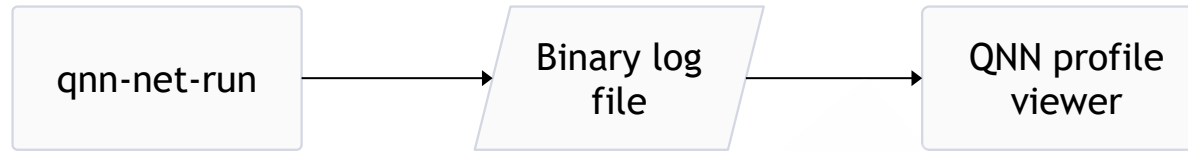
■ qnn-net-run configuration file (params.json)

```
{
  "backend_extensions" :
  {
    "shared_library_path": "<PATH-TO-SHARED-LIBRARY>",    // Shared library, for example: libLpaiNetRunExtensions.so
    "config_file_path" : "<PATH-TO-CONFIG-FILE>"          // LPAI backend config, for example: lpai_config.conf
  },
  ...
}
```

■ LPAI backend extension configuration file (lpai_config.conf)

```
{
  "lpai_backend": {
    "enable_hw_ver": "v6"
  },
  "lpai_graph": {
    "execute": {
      "fps": "number",
      "ftrt_ratio": "number",
      "client_type": "real_time/non_real_time",
      "affinity": "soft/hard",
      "core_selection": "number"
    }
  }
}
```

qnn-net-run: Basic profiling



```
qnn-net-run -profiling_level basic <OTHER NET RUN ARGUMENTS>
```

```
qnn-profile-viewer --input_log profile_log1,profiling_log2,...,profile_logx
```

```
Init Stats:
-----
NetRun: 94 us
Backend (QnnContext_createFromBinary): 722 us

Compose Graphs Stats:
-----
NetRun: 0 us

Finalize Stats:
-----
Graph 0 (CCT):
NetRun: 94 us
Backend (QnnGraph_finalize): 93 us

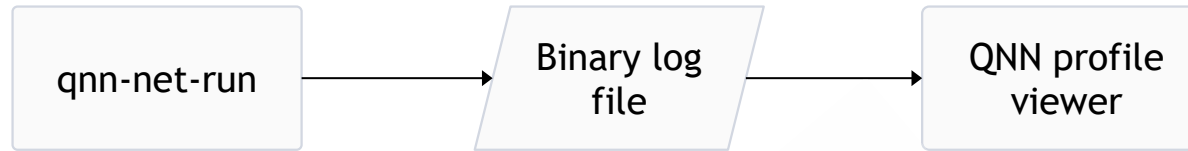
De-Init Stats:
-----
NetRun: 393 us
Backend (Context_Free): 6 us

Execute Stats (Overall):
-----
NetRun IPS (includes IO and misc. time): 19.3017 inf/sec
```

```
Execute Stats (Min):
-----
Total Inference Time:
-----
Graph 0 (CCT):
NetRun: 51761 us
Backend (QnnGraph_execute): 51757 us
IPC time: 78 us
Accelerator exec time: 51679 us

Execute Stats (Max):
-----
Total Inference Time:
-----
Graph 0 (CCT):
NetRun: 51761 us
Backend (QnnGraph_execute): 51757 us
IPC time: 78 us
Accelerator exec time: 51679 us
```

qnn-net-run: Detailed profiling



```
qnn-net-run -profiling_level detailed <other net run arguments>
```

```
qnn-profile-viewer --input_log profile_log1,profiling_log2,...,profile_logx
```

```
Execute Stats (Average):
-----
Total Inference Time:
-----
Graph 0 (CCT):
  NetRun: 51601 us
  Backend (QnnGraph_execute): 51549 us
  IPC time: 15 us
  Accelerator exec time: 51534 us
  Inference preparation time: 0 us
  layer Transpose_122 exec time: 1 us
  layer Transpose_241 exec time: 3 us
  layer Transpose_28 exec time: 4 us
  layer Transpose_285 exec time: 3 us
  layer Transpose_78 exec time: 5 us
  layer _layernorm_0 exec time: 42 us
  layer _layernorm_12 exec time: 41 us
```



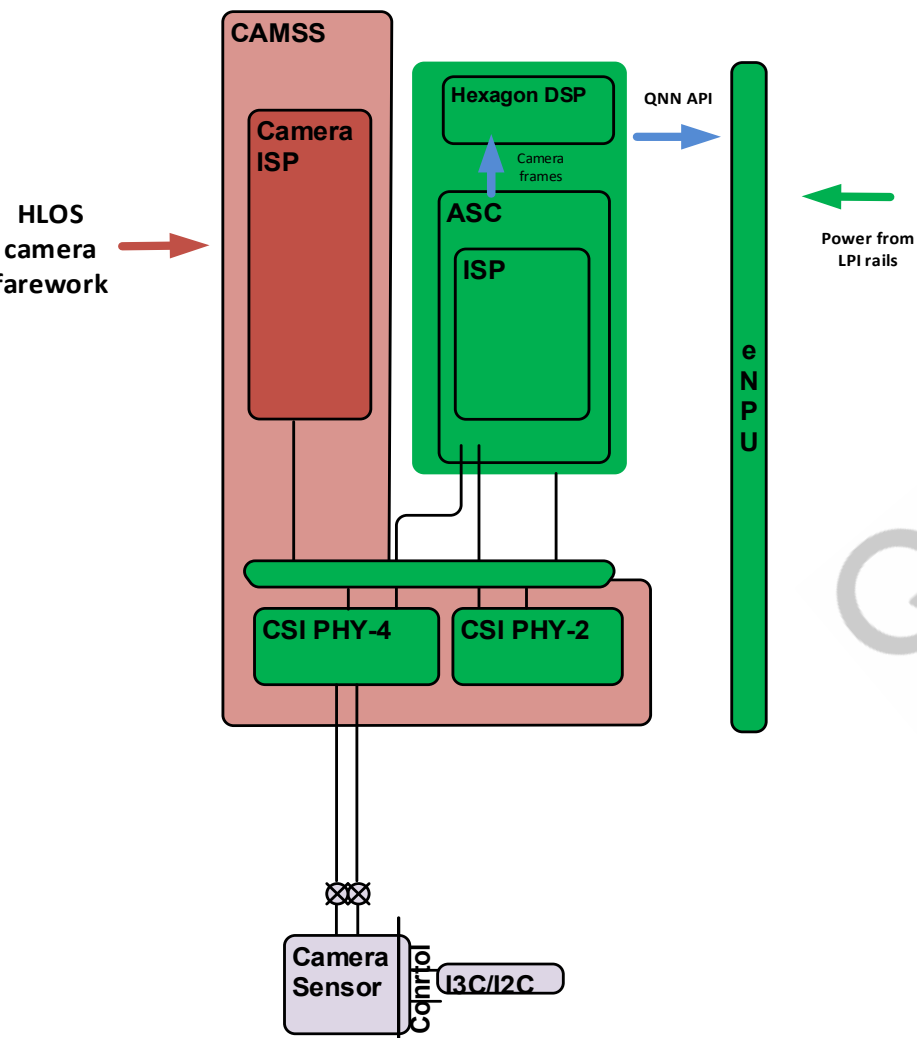
Section 6

Camera overview

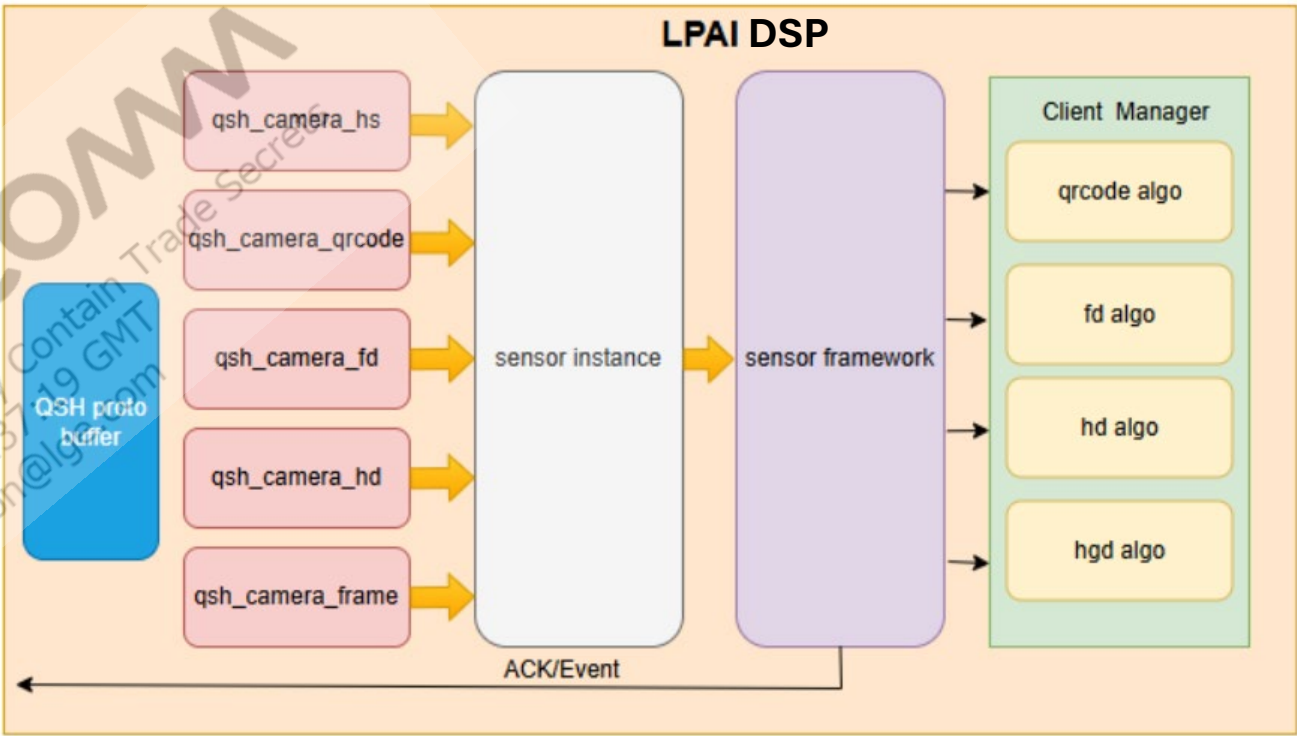
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Camera architecture

Camera ISP

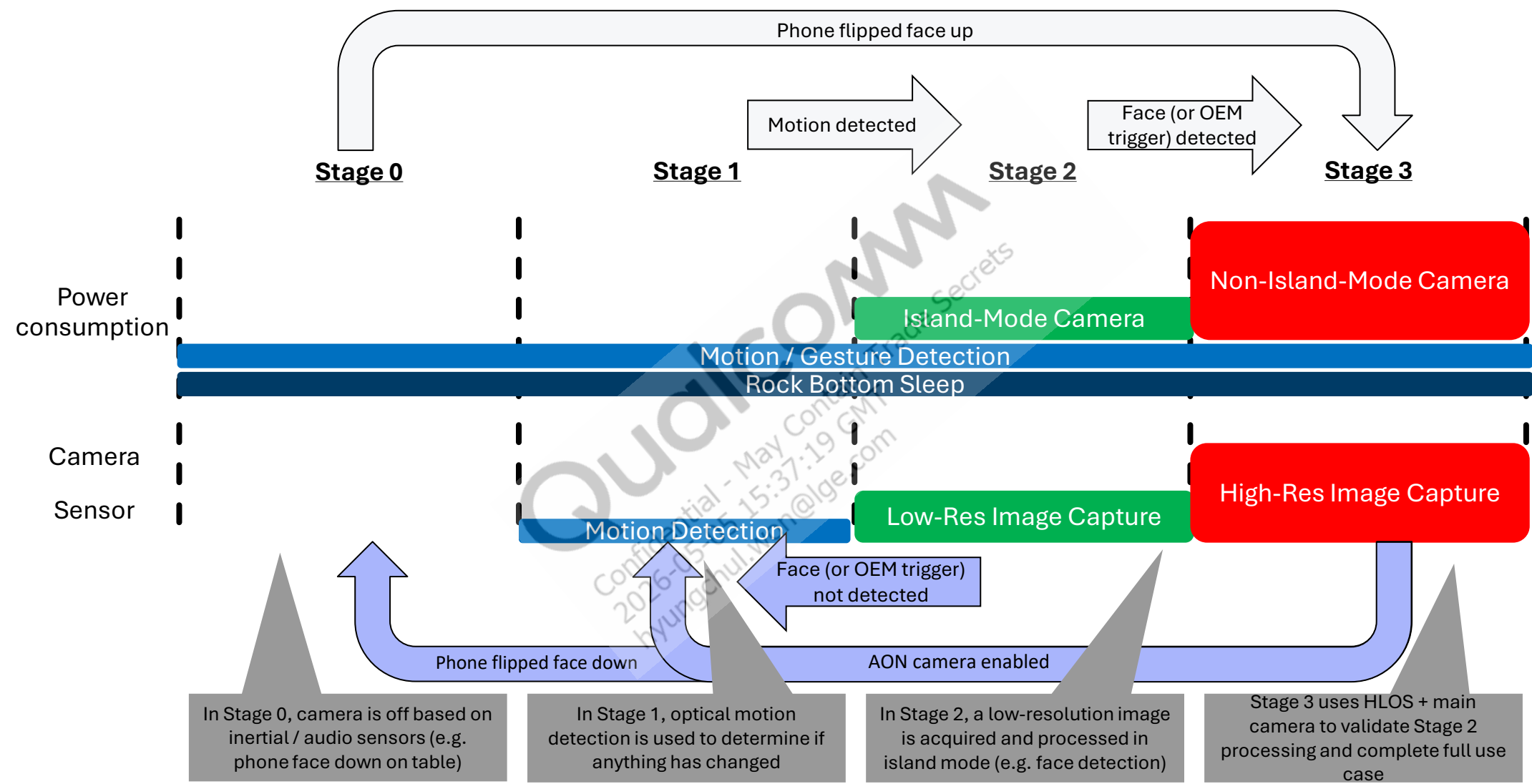


Main ASC Software component



NOTE: See the ISP documentation for more detailed information.

Always sensing camera (ASC) use case flow



Always sensing camera (ASC) software components

- Process the camera frame during island mode, wake up AP when needed
- A low-power, high-performance platform for OEMs on which to deploy their own Always-Sensing camera use cases such as Face detection or Motion detection
- Provide QSH camera OEM algo, which register the framebuffer, process the data with eNPU hardware
 - Sample code under `adsp_proc/qsh_drivers/qsh_camera_oem1`

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Always sensing camera (ASC) verification

- ASC verification is done using the unified sensor test app and SNS logs from the QXDM Professional™ Tool.

12:49

Unified Sensor Test App

USTA-TAB REGISTRY-TAB SENSORS-INF

Sensor: suid-QTI

On-Change: ccd_ttw-QTI

Rigid-Body-Type: camera_face_detect-QTI

SUID Low: proximity-amsAG

SUID High: proximity-amsAG

Scripts : ambient_light-amsAG

ReqMsgs: ambient_light-amsAG

Request: sar-Semtech

Suspend Config: sar-Semtech

Client Processor: sar-Semtech

Wakeup Delivery: humidity-Sensirion AG

Batch Spec: ambient_temperature-Sensirion AG

Batch Period (u-sec): sensor_temperature-STMicro

12:38

Unified Sensor Test App

USTA-TAB REGISTRY-TAB SENSORS-INF

Sensor: camera_face_detect-QTI

On-Change: False

Rigid-Body-Type: NA

SUID Low: e8e7f675544392a1

SUID High: a01f7e2dfc3bfae9

Scripts : Nothing is selected

ReqMsgs: qsh_camera_framework_island_subscribe-608

Request: Nothing is selected

Suspend Config: Nothing is selected

Client Processor: sns_std_client_processor-ssc

Wakeup Delivery: sns_client_delivery_wakeup

Batch Spec: Nothing is selected

Batch Period (u-sec):

Item List Config Match Config

Choose Item Type(s)

☐ Diag Requests

☐ Diag Responses

☐ Event Reports

☐ Log Packets

☐ Log Packets (OTA)

☒ Message Packets

☐ Strings

☐ QTraces

☐ QSH Events

☐ QSH Metrics

☐ Subsystem Dispatch Requests

☐ Subsystem Dispatch Responses

☐ SubsystemV2 Dispatch Requests

☐ SubsystemV2 Immediate Dispatch Responses

☐ SubsystemV2 Delayed Dispatch Responses

☒ Message Packets

☐ GNSS

☐ GNSS HLOS

☐ Blue Tooth

☐ WLAN

☐ QDSP6

☒ SNS

☐ FRAMEWORK

☐ PLATFORM

☒ SENSOR INT

☒ SENSOR EXT

☐ [00053/00] SNS/Low

☐ [00053/01] SNS/Medium

☐ [00053/02] SNS/High

☐ [00053/03] SNS/Error

☐ [00053/04] SNS/Fatal

☐ GSM

☐ LTE

Type	Time Stamp	Name	Summary
MSG	00:07:21.234483	SENSOR INT/...	[cam_sensor_framework_island.c 1260] Island Cnt: Enter 954 Exit 953
MSG	00:07:21.220000	SENSOR INT/...	[cam_core_wrappers_utmter_island.c 114] cam_utmter_stop (timer = C3CA0009)
MSG	00:07:21.220000	SENSOR INT/Low	[cam_sensor_framework_island.c 1162] transition_to_stage_1
MSG	00:07:21.220000	SENSOR INT/...	[qsh_camera_sensor_island.c 769] vote_tcm_lpr_cb - ALLOW_LPR
MSG	00:07:21.220000	SENSOR INT/...	[cam_core_wrappers_utmter_island.c 101] cam_utmter_start: (timer = C3CA0009) 500 ms
MSG	00:07:21.220000	SENSOR INT/...	[cam_core_wrappers_gpio_island.c 101] cam_gpio_write - pin_num (18) gpio_state (0)
MSG	00:07:21.723001	SENSOR INT/...	[cam_sensor_framework_island.c 1944] alsValue: 0xd77
MSG	00:07:21.723303	SENSOR INT/Low	[csid_program_island.c 41] cam_aon_csid_reset
MSG	00:07:21.723485	SENSOR INT/...	[phy_program_island.c 44] cam_phy_program: About to poll for refgen ready
MSG	00:07:21.723489	SENSOR INT/...	[phy_program_island.c 47] cam_phy_program: refgen ready
MSG	00:07:21.720000	SENSOR INT/Low	[cam_sensor_framework_island.c 1271] transition_to_stage_2
MSG	00:07:21.720000	SENSOR INT/...	[qsh_camera_sensor_island.c 614] i2c_rw_cb - is_write = 1 num_vectors = 1 rc = 0
MSG	00:07:21.795254	SENSOR INT/...	[cam_sensor_framework_island.c 1260] Framework state moved to FW_STATE_STAGE_1
MSG	00:07:21.795260	SENSOR INT/...	[cam_sensor_framework_island.c 1260] Island Cnt: Enter 965 Exit 964
MSG	00:07:21.725000	SENSOR INT/...	[cam_core_wrappers_gpio_island.c 101] cam_gpio_write - pin_num (18) gpio_state (0)
MSG	00:07:21.756250	SENSOR INT/...	[qsh_camera_sensor_island.c 614] i2c_rw_cb - is_write = 1 num_vectors = 39 rc = 0
MSG	00:07:21.756250	SENSOR INT/Low	[cam_isp_driver_island.c 141] cam_isp_driver_start
MSG	00:07:21.756250	SENSOR EXT/...	[caonaec_island.cpp 297] Exp table knee: Gain: 2.000000 Exp Time:20000.000000
MSG	00:07:21.756250	SENSOR EXT/...	[caonaec_island.cpp 335] AONAEC Arbitration: Before AntiBand-Index:280.000000 G:3.144208 ET:8333333.00000
MSG	00:07:21.756250	SENSOR EXT/...	[caonaec_island.cpp 422] Antibanding not performed: ET: 8333333.000000 nsec less than antibanding time p
MSG	00:07:21.756250	SENSOR EXT/...	[caonaec_island.cpp 407] AONAEC StartUp: Mode:1 StartUp Index 280.000000, ALS exposure:3447.000000 Ind
MSG	00:07:21.756250	SENSOR INT/...	[cam_sensor_framework_island.c 1856] isp_cb_aec_ready: expTime 8333333 gain 3 digitalGain 1



Section 7

References

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Documents and Standards

Documents	
Title	Number
Qualcomm Technologies, Inc.	
<i>AudioReach Signal Processing Framework Overview</i>	80-VN500-3
<i>QACT v8.x.x User Guide</i>	80-VM407-18
<i>SEE Sensors Deep Dive</i>	80-P9301-35
<i>Sensors Execution Environment Client API</i>	80-P9301-36
<i>Adding a Custom Sensors Algorithm with Sensors Execution Environment (SEE)</i>	80-P9301-67
<i>Unified Sensor Test Application (USTA) User Guide</i>	80-p9301-85
<i>Adding a Custom Sensors Algorithm with Qualcomm Sensing Hub (QSH)</i>	80-pv345-111
<i>Sensors Execution Environment (SEE) Sensors Bringup Guide</i>	80-PV345-103
<i>LPAI DSP Stability Debugging Guide</i>	80-40939-76
<i>Context Change Detector (CCD) 6.0 and Context Engine Configuration Tool (CECT) 6.0 User Guide</i>	80-60672-3
<i>eNPU6 Design Guidelines</i>	80-37209-5
<i>Qualcomm Sensing Hub (QSH) Overview</i>	80-40939-17
<i>CAPi Custom Module Integration User guide</i>	80-VN500-17

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